

Bachelor in **Applied Data Science**

FEATURE BEHAVIOR ANALYSIS FOR DDoS DETECTION ACROSS IOT AND IIOT NETWORKS

ÉRICA BERG

SUBMITTED IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE
AWARD OF THE DEGREE OF BACHELOR IN **APPLIED DATA SCIENCE**

SUPERVISOR

Bertram Haskins

Noroff University College
Norway
Apr, 2026

Abstract

The rapid expansion of the Internet of Things (IoT) and Industrial Internet of Things (IIoT) has increased the number of Distributed Denial of Service (DDoS) attacks. Machine learning approaches have been widely applied for intrusion detection; however, feature behavior is often analyzed within a single network context, limiting generalization across domains. This research provide a cross-domain analysis of feature relevance in IoT and IIoT environments by combining quantitative methodology, statistical analysis, and insights from the literature. Two different datasets with distincs structures were analyzed to evaluate statistical behavior, correlation, discriminative power, and feature importance. Next, reduced datasets derived from the primary ones were created gathering the most relevant features for distinguishing between traffic classes. Findings were then validated using Random Forest and Support Vector Machine models. Results indicate that traffic intensity is a primary indicator of DDoS attacks in both environments, while structural features also contribute significantly to detection performance. In contrast, temporal features showed better performance in IoT than in IIoT scenarios. The findings suggest that feature behavior depends not only on the network environment, but also on their representation. Overall, this study provides evidence-based insights for cross-domain feature selection in DDoS detection.

Keywords: *IoT, IIoT, DDoS detection, feature selection, machine learning*

Acknowledgements

I offer my deepest gratitude to Exu for keeping my mind strong and my paths open throughout this challenging journey.

I also want to thank my beloved husband, Peter Berg, who encouraged me to pursue this degree and held my hand every single day. He constantly inspires me with his intelligence and companionship. To my daughter, Silje Berg, my greatest source of inspiration, who is a source of endless energy, joy, and curiosity for learning — I love you both.

I am deeply grateful to Professor Bertram Haskins for his professionalism as a teacher and supervisor, for his generosity in sharing his knowledge so sincerely, and for his sensitivity in keeping us students motivated and confident during the most challenging moments of this journey.

I would also like to thank my colleague, Johannes Framnes, for being the comic relief throughout the past year. I am certain that our meme battles made this journey much lighter.

I also thank my friends for understanding and respecting my absence, especially Clara Caetano, who, even from a distance, is present in my life every day, and my friend Andrea Mantovani, who constantly reminds me of the importance of education for independence, the courage to start over, and the strength of friendship.

To my family, the foundation of my life.

Laroyê Exu.

Mandatory Declaration

Declarations

The individual student is responsible for familiarising themselves with the rules and regulations regarding the use of sources, generated text and academic misconduct. Failure to declare **does not** release the student from their responsibility, and will result in an automatic failure of the submission.

1.	I hereby declare that the submission made is my own work, and that I have not used other sources other than as referenced and cited correctly, or received help other than what is specifically acknowledged.	Yes
2.	I further declare that this submission: • Has not been used for another exam in another course at Noroff University College, at another department/university/college at home or abroad. • Does not refer to or make use of the work of others without acknowledgement. • Does not refer to my own previous work unless stated. • Has all the references given in the bibliography. • Is not a copy, or duplicate of someone else's work or answer. • Is not generated using AI generation tools.	Yes
3.	I am aware that a breach of any of the above is to be regarded as cheating and may result in cancellation of the exam and exclusion from universities and colleges in Norway, cf. University and College Act §§12-4, and NUC Regulations on examinations §§4-5.	Yes
4.	I am aware that all components of this assignments may be checked for plagiarism and other forms of academic misconduct.	Yes
5.	I hereby acknowledge that I have been taught the appropriate ways to use the work of other researchers. I undertake to paraphrase, cite, and reference according to the acceptable academic practices, in accordance with the rules and guidelines, as taught.	Yes
6.	I am aware that Noroff University College will process all cases where cheating is suspected in accordance with the college's guidelines.	Yes

Publication Agreement

Authorisation for electronic publication of the thesis: Through submission you are accepting that Noroff University College has a perpetual, and royalty free right to retain a copy of work for its own internal use, and has the right to make work publicly available - considering any restrictions to publication.

Name: Érica Berg
Date: April 30, 2026

Place: Oslo

Updated 4/9/2025

Contents

1	Introduction	1
1.1	Introduction	1
1.2	Background and Related Work	2
1.3	Problem Statement	3
1.4	Research Objectives	4
1.5	Research Methodology	4
1.6	Primary Data Set	5
1.7	Scope and Limits	5
1.8	Document Structure	5
2	Literature Review	7
2.1	Internet of Things and Industrial Internet of Things	8
2.2	IoT Architecture	9
2.3	IloT Architecture	12
2.4	Attacks Taxonomy, Detection and Mitigation	14
2.4.1	Recon	14
2.4.2	Brute Force	15
2.4.3	Spoofing	15
2.4.4	Web-Based Attacks	15
2.4.5	DoS	16
2.4.6	Mirai (Case study from 2016)	16
2.4.7	DDoS	17
2.5	Machine Learning for IDS	18
2.6	Research Gaps	19
3	Methodology	21
3.1	General methodology overview	21
3.2	Research Method: Literature Review	22
3.3	Research Method: Data Analysis	22
3.3.1	Dataset Selection	22
3.3.2	Exploratory Data Analysis	24
3.3.3	Categorical Feature Analysis	24
3.3.4	Correlation Analysis	25
3.3.5	Protocol and Group Feature Analysis	25
3.3.6	Discriminative Power Analysis (RBC)	26
3.3.7	Feature-Level Analysis and Dataset Reduction	26
3.4	Research Method: Experimentation	27
3.4.1	Evaluation	27
3.4.2	Cross-domain analysis	28
3.5	Evidence-based guidelines for feature selection in IoT-IloT networks	28

4	Implementation and Results	29
4.1	Implementation	30
4.1.1	IoT Data Preparation	30
4.1.2	IIoT Data Preparation	31
4.1.3	Statistical Analysis Execution	33
4.1.4	Feature Selection and Dataset Reduction	33
4.1.5	Selected Features	34
4.1.6	Baseline Model Implementation	35
4.1.7	Evaluation Procedure	36
4.2	Statistical and Analytical Results	36
4.2.1	IoT	36
4.2.2	Industrial IoT	37
4.3	Destination Port	38
4.3.1	IoT	38
4.3.2	Industrial IoT	38
4.4	Feature Correlation Results	39
4.4.1	IoT	39
4.4.2	Industrial IoT	40
4.5	Feature Behavior Across Structural Categories	41
4.5.1	IoT	41
4.5.2	Industrial IoT	42
4.6	Discriminative Power Results	43
4.6.1	IoT	43
4.6.2	Industrial IoT	44
4.7	Feature Selection	45
4.7.1	IoT Features	45
4.7.2	Industrial IoT Features	50
4.8	Model Performance for Feature Selection Validation	56
4.9	Discussion	57
4.10	Guidelines for Feature Selection	59
4.10.1	Guideline 1: Traffic Intensity as a Primary Indicator	59
4.10.2	Guideline 2: Packet Size and Length as Structural Indicators	59
4.10.3	Guideline 3: Domain-Specific Behavior of Temporal Features	60
4.10.4	Guideline 4: Destination Port Handling Across Domains	60
4.10.5	Guideline 5: Importance of Dataset-Specific Validation	60
5	Conclusion	62
5.1	Introduction	62
5.2	Summary of Research	62
5.3	Research Objectives	63
5.4	Research Contribution	64
5.5	Future Work	64
	References	66
A	Sample Appendix	73

List of Figures

4.1	Top-15 IoT features ranked by standard deviation	37
4.2	Mean vs Median - top-10 most variable IoT features	37
4.3	IoT Correlation Heatmap - Top-10 by Standard Deviation	40
4.4	IoT Correlation Heatmap - Top-10 by Standard Deviation	41
4.5	Standard deviation - Difference between TCP - UDP	42
4.6	Discriminative Power Across Groups	43

List of Tables

4.1	Numerical features used in the CIC IoT-DIAD 2024 dataset	30
4.2	Numerical features used in the CIC IIoT 2025 dataset	32
4.3	Selected features for the IoT dataset	34
4.4	IIoT Reduced Dataset	35
4.5	Performance values for IoT and IIoT models.	57

1

Introduction

1.1 Introduction

The Internet of Things (IoT) is a concept that is becoming increasingly present in modern technology. Its main idea is to enable everyday objects to be connected, allowing them to interact with each other through the Internet (Atzori et al., 2010). Feroz Khan and G (2021) highlights the concept's diversity by stating that objects can be anything, such as a person with a blood pressure monitoring implant, an animal with a transponder and a car connected with sensors. These objects must be assigned an Internet Protocol (IP), allowing data to be transferred over a network. This connectivity has expanded the application of IoT in several areas, such as smart homes, healthcare, environmental monitoring, and industrial systems (Kayode Saheed et al., 2022). In industrial environments, this evolution is known as the Industrial Internet of Things (IIoT), or Industry 4.0, and it is the application of the IoT concept to the industrial sector, where monitoring and production processes involve a significant number of networked embedded sensing devices (Peter et al., 2023).

Although the incorporation of these technologies into our daily life brings many benefits, their rapid increase has become challenging, especially in terms of cybersecurity (M. H. Ali et al., 2022). In this sense, Casola et al. (2019) highlights that physical objects transformed into internet hosts are targeted by attackers, threatening users' privacy and security. Among these threats, Distributed Denial of Service (DDoS) attacks have become especially relevant and heavily used by criminals in recent years.

Intrusion Detection Systems (IDS) have been explored to mitigate these threats, where machine learning techniques have been widely used and provided satisfactory results. Defining an effective set of features is essential in this context, as it directly impacts model performance, interpretability, and generalization (Jo et al., 2019).

Existing studies often explore IoT and IIoT separately, limiting the analysis of feature behavior across domains. In addition, automated feature selection is applied to many approaches, which may result in dataset-dependent models. As a result, there is a lack of cross-domain analysis that investigates patterns of feature behavior identified in IoT and IIoT, limiting the generalization of machine learning models in DDoS detection.

This study provides a cross-domain analysis of feature behavior in IoT and IIoT environments, combining statistical analysis, insights from the literature, and experimental validation using machine learning models. Findings further support the development of guidelines for feature selection, aiming to improve the interpretability and generalization of DDoS detection across different environments.

1.2 Background and Related Work

IoT can be described as a collection of physical devices, also known as smart devices, that connect to a network and are able to exchange data in real time. These devices are widely applied in both domestic and industrial environments, that go to smart homes to large-scale monitoring systems (Aly et al., 2019). The expansion of IoT to industrial segments is known as Industrial IoT (IIoT), where these interconnected devices monitor and control industrial sector in an automatized process (Peter et al., 2023). While both domains rely on device connectivity and data exchange, the IIoT environment is more machine-to-machine oriented and usually demands higher levels of reliability, scalability, and real-time communication and privacy (Sisinni et al., 2018).

Both IoT and IIoT support great advantages; however, many security challenges are associated with them. In IoT, many devices are resource-constrained, with limited processing power, memory, and energy capacity, posing as a limitation to implementing robust security mechanisms (Kayode Saheed et al., 2022). Additionally, these devices are highly heterogeneous, exploring different hardware, software, protocols, and communication patterns (Sahu et al., 2021). Similar challenges are also present in IIoT, where connectivity on a large scale and modern systems operating with real-time communication reinforce the need for data reliability, availability, and security. Therefore, issues such as interoperability, scalability, and device security must be addressed in both domains to allow safe and efficient connection (Xu et al., 2014).

The combination of limited resources and devices heterogeneity makes these environments attractive to cyberattacks. Among several types of attacks, Distributed Denial-of-Service

(DDoS) attacks have been lately favored by attackers, in which compromised devices generate large volume of malicious traffic, interrupting the regular distribution of service. Traditional defense mechanisms fail to effectively secure IoT and IIoT environments, resulting in constant study and proposal of intelligent detection mechanisms approaches to identify malicious activity in network traffic. In this context, machine learning techniques have been widely explored, as they enable large-scale data analysis and capture variations in benign and malicious behavior (Casola et al., 2019).

Machine learning approaches have been widely used to address security issues in IoT and IIoT environments, since these techniques thrive when traditional mechanisms defenses implementation is limited (Aly et al., 2019; Mirani et al., 2022). Many IoT devices are memory and power limited, have reduced processing capacity, and traditional cryptographic security mechanisms cannot always be directly implemented, exposing these devices to threats such as eavesdropping, data leakage, and denial-of-service attacks. Therefore, machine learning-based intrusion detection systems (IDS) have emerged as an effective alternative to continuously monitor abnormal patterns in network traffic, enabling their implementation under constraints such as low latency, while providing adaptive and scalable security solutions for both IoT and IIoT environments (Kayode Saheed et al., 2022).

Many solutions are being proposed to detect DDoS attack in IoT and IIoT devices, however, most studies focus on isolated environments or datasets, limiting the understanding of features behavior across domains, especially considering the distinct communication patterns between IoT and IIoT. As result, the generalization of machine learning approaches might be reduced, since detection models might be dependent on a specific environment or dataset. Therefore, investigating feature behavior across domains is fundamental to identify features that are dependent on a specific context, directly contributing to the development of more adaptable and reliable intrusion detection systems.

1.3 Problem Statement

The increasing expansion of IoT networks introduced several benefits and different possibilities of accessibility to its users, but also brought significant security challenges (Roopak et al., 2020). Furthermore, Chaudhary and Mishra (2023) describe Industrial IoT as an extension of IoT applied to the industrial sector.

Studies reveal that Distributed Denial of Service (DDoS) has impacted these sectors (Casola et al., 2019), mainly because they involve vulnerable devices, the scalability of attacks, and the ease of execution. These attacks compromise multiple devices, interrupting services, and causing the denial of service to legitimate users.

Several studies have been conducted to improve the identification of malicious activities, and feature selection techniques have been widely explored to improve detection performance.

However, studies are limited to exploring the domains separately or performing automated feature selection. With this, there may be limitations in the generalization of the results, especially when trying to understand the behavior of the features in different domains. Therefore, the problem statement addressed by this study is **DDoS anomaly detection lacks cross-domain analysis of feature behavior**, highlighting a critical gap in current research.

1.4 Research Objectives

The primary objective of this research is to **develop practical guidelines for identifying and selecting relevant features for DDoS detection across IoT and IIoT domains**, especially regarding to feature behavior, statistical and theoretical relevance, and also the discriminative power of these features.

To achieve this objective, secondary objectives are defined as follows:

- **SO1:** To perform a comparative feature behavior analysis, with special attention to the unique characteristics of each domain.
- **SO2:** To identify feature relevance, redundancy and protocol-dependent and domain-dependent behavior, in order identify a set of applicable features for DDoS detection.
- **SO3:** To perform an experimental validation of the findings using baseline machine learning models, assessing whether the reduced feature sets are sufficient to support reliable DDoS detection across both domains.

1.5 Research Methodology

This study adopts a quantitative and data driven-research approach to understand traffic behavior, especially the differentiation between benign and DDoS network traffic across IoT and Industrial IoT environments. Two different datasets are used in this study and not only they are from different environments but also have different types of data representation: flow-based and window based. In addition to the statistical inferences from the datasets, existing literature is used to support feature selection and relevance.

For each full dataset, two reduced versions are created with best features selected to understand network patterns. In addition, baseline machine-learning models are applied: Random Forest and Support Vector Machine (SVM), to confirm whether the features identified in previous stages improve consistency when detecting DDoS traffic across both IoT and IIoT (Géron, 2018). Model performance is evaluated using standard evaluation metrics, including accuracy, precision recall and F-score.

Finally, this project elaborates evidence-based guidelines for feature selection for DDoS attack identification in IoT and IIoT environments.

1.6 Primary Data Set

This research is based on two pre-existing datasets: CIC IoT DIAD 2024 (Rabbani et al., 2025) and CIC IIoT 2025 (Firouzi et al., 2025). The datasets provide benign and DDoS network traffic samples collected from IoT and IIoT environments, respectively.

These datasets provide reliable data for investigating feature behavior across different network domains while supporting statistical analysis and machine learning validation.

1.7 Scope and Limits

This study focuses on the identification of feature patterns that distinguish between benign and DDoS traffic, which are limited to the analysis of IoT and IIoT datasets. The study aims to understand DDoS traffic by analyzing feature behavior, applying statistical methods, and validating the results using machine learning models.

This study is also limited to datasets collected in a single time period, therefore does not support cross-year evaluation.

Machine learning models are used exclusively to validate the theory developed in this project, and deep learning approaches or more advanced machine learning models are not addressed. In addition, no feature engineering is performed and the study is limited to the features available in the datasets.

Finally, this study solely focuses on DDoS attack, and other attacks eventually identified in the datasets are not considered.

1.8 Document Structure

This study is organized into well-structured chapters that first present a theoretical background covering the different topics involved, listing the objectives, research methodology, and definition of the scope and limitations. It is structured as the following:

- Chapter 2 - Presents the foundation of the theoretical background and related studies understand IoT and IIoT traffic. It introduces the network architecture for both domains, as well attack taxonomy, detection, and mitigation approaches.
- Chapter 3 - Describes the research methodology, specifying how each topic aligns to the research objectives, providing the reasoning behind the selected methods.
- Chapter 4 - Describes implementation process and the results obtained from it, including feature behavior analysis, statistical observations, model evaluation and experimental validation.

- Chapter 5 - Summarizes the main findings and discusses their implications. At the end, future work recommendations are given based on the limitations observed during the analysis.

2

Literature Review

The increasing growth of IoT devices is justified by many factors, such as demand for ease, security and low energy cost. Sinha (2025) reported a 14% growth in IoT devices just in 2025 and estimates that there will be over 39 billion devices connected to the internet by 2030. Combined with the rapid growth, McKinsey (2018) estimates that the economic impact in 2025 can reach \$11.1 trillion. However, this same growth and improvement across multiple domains has also introduced unprecedented security challenges (Kayode Saheed et al., 2022).

IoT devices operate with limited resources, making proper security design challenging. In addition, the concept of heterogeneity – different devices, operating systems, protocols, and communication patterns – is a key point in various studies regarding IoT security challenges. Casola et al. (2019) highlights that the combination of different software and hardware represents a limitation for a single solution capable of protecting the entire IoT ecosystem.

Security vulnerabilities are also reinforced by weak factory-set passwords and insecure protocols that rarely receive updates (Pakmehr et al., 2024). Additionally, Gu et al. (2020) highlights that some IoT devices present weak or nonexistent encryption capabilities. As a result, IoT has become an attractive target for cyberattacks, particularly Distributed Denial-of-Services, where security vulnerabilities in this collection of heterogeneous devices are exploited and infected devices generate massive traffic floods (Pakmehr et al., 2024).

Over the years, research has proposed and explored different strategies to mitigate these risks, such as the adoption of lightweight encryption that supports secure authentication,

anomaly-based intrusion detection, and Machine Learning-based detection. However, some topics remain a challenge, such as feature selection and scalability, and they pose a limitation for some research, especially when the proposed models cannot be leveraged in other datasets or lose their strength over time.

Therefore, this literature review aims to gather and investigate past studies that have addressed IoT and IIoT security, focusing on those that have explored DDoS and machine learning techniques. It is expected to collect knowledge on their architecture and protocol-level characteristics, explore other common cyberattacks and their existing detection strategies, and research gaps. Also, the literature review highlights these underexplored areas: lack of cross-dataset, cross-year evaluation, and the insufficiency of research that explores the technical differences between IoT and IIoT networks. Given the dynamic nature of IoT traffic and systematic evolution of cyberattacks, it becomes essential that emerging studies correlate machine learning approaches to concept drift, so systems proposed remain robust over time.

The structure of this investigation begins with fundamental concepts of IoT and IIoT and their respective architecture protocols, highlighting the technical underlying differences among them. The literature review then examines security challenges and most recurring cyberattacks on these networks, followed by an analysis of how previous research presented solutions with detection and mitigation strategies, particularly those with machine learning-based approaches. Finally, this section introduces the discussion on engineering and traffic behavior, which constitutes the initial yet highly complex stages of machine learning methods that operate across such heterogeneous environments.

2.1 Internet of Things and Industrial Internet of Things

IoT refers to devices that are interconnected and allow interaction between machine-to-machine (M2M), machine-to-man, and man-to-man (Choudhary, 2024). Overall, IoT devices must be able to respond to the physical world without human intervention (Kayode Saheed et al., 2022).

IoT traces back to the 1980s, with a Coca-Cola vending machine that verified the drink availability and temperature before performing the purchase. However, the concept of the IoT came only in 1999 with Kevin Ashton, MIT's Executive Director of Auto-ID Labs. Its creation derived from a logistical issue, not the internet. The main idea was the development of a technology that could connect objects to the computer, and according to Ashton, if all objects could be "tagged" and digitally identified, computers could track and efficiently organize them. Radio Frequency Identification was a key point to allowing things to communicate autonomously through data (D. Foote, n.d.).

However, in 2012, when major internet provider companies agreed to adopt the Internet Protocol version 6 (IPv6), the objects connected to the internet massively increased. From

this milestone, objects that somehow operate with data transferring would have an IP address (Cisco, n.d.). Nowadays, it is estimated that non-IoT devices represent 1/3 of the IoT devices connected to the internet (Petroc, 2025).

The rapid expansion of IoT has led to its integration into several domains that goes from industry to everyday life. According to Kumar Mishra and Raghuvanshi (2024), IoT applications include predictive maintenance, financial services, smart cities, agriculture, and environmental management.

Predictive maintenance is widely applied in industrial environments, where real-time data analysis can predict device and machinery failures before they occur. Sensors monitoring parameters such as temperature, pressure, vibration, and humidity collect data and send data to data warehouses or cloud-based platforms that utilize machine learning algorithms to analyze patterns and identify potential failures or malfunctions, resulting in downtime, reducing operational costs and optimizing asset performance and lifespan (Latorre et al., 2025).

The IoT is also changing the financial services sector. With smart payment systems, credit/debit cards are now being replaced for wearable devices, such as smart watches or other IoT-enabled devices. Financial institutions are also benefiting from asset tracking, risk assessment, and fraud detection (Al Mamun et al., 2023).

IoT technology is also a key component for developing smart cities, since interconnected sensors and devices work with a flow of collected and exchanged data that improve decision-making, automate processes, and enhance urban services. According to the author, IoT in smart cities encounters multiple domains, such as transportation, energy management and public safety. For instance, in smart transportation, IoT sensors collect real-time, and monitor traffic conditions, thereby, improving urban mobility (Rehan, 2023).

This expansion of IoT to industrial segments represents the fourth Industrial Revolution, driven by device interconnectivity, automation, decision making power, machine learning and real-time data (Munirathinam, 2020). Although the foundational networking technologies are the same, they differ in many other characteristics, mostly justified by the fact that IIoT must provide scalability, reliability and real-time communication (Leng et al., 2022).

Therefore, the rapid growth of IoT devices combined with the vast areas that apply IoT concepts demonstrate not only technological and economic relevance but also highlights the importance of understanding their communication protocols, addressing security and privacy issues in IoT/IIoT networks.

2.2 IoT Architecture

Despite the constant creation and connection of IoT devices to the internet, IoT network remains without a standard IoT design. Therefore, while literature has not precisely defined the

the IoT architecture layers, a great portion of the research has presented IoT's architecture in 3 layers as follows(Sadique et al., 2018), (Pakmehr et al., 2024):

The Perception Layer is the foundation layer in IoT architecture and is responsible for providing support for data collection from the surroundings through physical objects, such as sensors and actuators, detecting and realizing the first step of the information processing. Sensors differ in their types since they must serve a specific purpose, such as measuring humidity, temperature, and pressure (Bhajantri & S., 2021). Usually, they are small in size, capacity-constrained, and can operate using minimal processing power, often relying on microcontrollers without any embedded operating system. The layer supports various devices because they are integrated into the IoT according to the device's purpose (Domínguez-Bolaño et al., 2022). According to Sadique et al. (2018), most of the connected nodes consist of RFID (Radio-frequency identification) tags or wireless sensors. However, TCP/IP (Transport control protocol/Internet protocol) remains the dominant protocol used for internet communication. There are many non-IP IoT devices or devices that rely on short-range communication that operates below the TCP/IP layers. As highlighted by Gerber and Romeo (2020) and Zhao and Ge (2013), most technologies in this layer includes RFID, Zigbee.

Although the architectural layer in this project is divided into 3 layers, some studies have shared the same division understanding as OSI models, adding a Data Link Layer between the Perception and Network Layer.

Although this project adopts the architecture divided into 3 layers, some studies have aligned their architectural division similar to the OSI model, adding the Data Link Layer between the Perception and Network Layer. In this sense, Choudhary (2024) states that the Data Link Layer must fulfill all requirements for hop-to-hop communication. Like the OSI model, this layer includes the MAC sublayer that enhances network efficiency:

Data Link Layer support reliable data transmission via error detection and correction mechanisms which ensure that data corruption during transmission is minimized, leading to reliable communication, low power consumption technologies through the use of Zigbee, LoRa, Bluetooth which is crucial for IoT devices operating on limited power resources, access control provided by MAC sublayer to efficiently manages shared communication mediums by avoiding collisions and transmit data seamlessly. This improves overall network efficiency. (p. 12-13)

Additionally, Karimi and Shaefer (2025) list IoT wireless Physical and Link Layer protocols which include IEEE802.15.4, Zigbee, Wi-Fi, Bluetooth, BLE (Bluetooth low power) and LoRaWan.

While the previous layer works by collecting and encapsulating data into frames, **Network Layer** is responsible for enabling data transmission across other IoT devices and sending these packets across multiple nodes, networks, and gateways. This layer supports a variety of protocols, the most common are IPv4, IPv6, and 6LoWPAN, which work below the

TCP/IP protocol. The first, IPv4, handles 32-bit addresses, which converts to approximately 4.3 billion unique addresses. This represents a highly inferior amount of IP addresses compared to the exponential growth of IoT devices. IPv6 was introduced to solve the address exhaustion problem with a 128-bit address space. However, what came as a solution had also become a challenge for IoT networks. Compared to IPv4, IPv6 has larger headers, and many low-powered IoT devices struggle to fit them into the data link layer frames (IPv4Mall, 2024).

Internet Engineering Task Force (IETF) proposed a solution that has largely been adopted to IoTs. The IPv6 over Low-Power Wireless Personal Area Networks (6LoWPAN) compresses and fragments IPv6 headers so they can operate on small low-power IoT devices (Salman & Jain, 2019).

Despite the importance of this layer in scalability, interoperability and being the operator of data transfer across different protocols, this layer has inherent resource constraints and makes IoT networks vulnerable to data interception and unauthorized access, in this sense, some studies, such as Talal and Zagrouba (2021) have considered this to be one of the most vulnerable layers.

Finally, the **Application Layer** is where all data collected and transmitted by the lower layers is returned to its clients through applications protocols (Adam et al., 2024). According to Zhao and Ge (2013), the Application Layer is the architecture level in IoT that holds more technological complexity. At this layer, the information gathered from sensors and RFID and transmitted by lower layers is finally converted into specific services to the end user. The authors list some key technologies to be applied at this layer, such as Cloud Platforms, Application Programming Interfaces, Artificial Intelligence, Machine Learning, and Data Analytics tools. Although these technologies belong to the Application Layer, they are not executed at the IoT device itself, due to the fact that they are constrained resources.

Instead, the high technology from this layer allows those operations to happen at external infrastructure, such as cloud services, fog nodes, or edge computing platforms, which provide computational and analytical resources for large-scale and real-time decision making. This is possible because of the protocols that act in this layer are constantly being created and adapted to support the growth of the IoT device network (HaddadPajouh et al., 2021). Choudhary (2024) lists Message Queue Telemetry Transport (MQTT), Constraint Application Protocol (CoAP), Advanced Message Queuing Protocol (AMQP), Extensible Messaging and Presence Protocol (XMPP), Data-Distribution Service protocol (DDS), TCP/IP, and UDP/IP.

MQTT is mostly used for small IoT devices operating under low bandwidth, unstable network conditions, and limited processing power, enabling efficient machine-to-machine communication. Bibi et al. (2021) also surveyed application protocols, detailing their functionalities, and provided a comparative analysis. For instance, the same highly resource-constrained

IoT devices often rely on CoAP, which is a lightweight REST-based transfer protocol that operates under User Datagram Protocol (UDP) reducing overhead while supporting multicast communication, which is suitable for low-powered networks. These limited memory devices are then benefited from CoAP, where a simplified subset of HTTP functionalities is used to perform efficient communication. In other words, CoAP was designed to imitate HTTP in a lightweight mode, suitable for IoT networks. However, the author also highlights that despite the advantages of CoAP protocols, the combination of UDP with CoAP results in message unreliability, because even if these protocols ensure that a message was received, there is no guarantee whether the content was fully received or properly decoded.

XMPP is an XML-based messaging protocol designed for real-time communication that allows devices to exchange data with proper authentication, access control, and encryption. XMPP architecture does not depend on a single central server, avoiding overload allowing scalability, while it supports presence detection and publish/subscribe messaging making it suitable for IoT applications that require coordination between multiple devices. However, there is no official support for this protocol, which makes data transferred repeated, causing great overhead.

Nevertheless, these applications can be applied in several domains, such as smart cities. Smart homes, smart grids, and healthcare, making security concerns unique for each environment. Additionally, since this layer deals directly with end users in terms of services, security threats often target data theft, malware propagation, and botnet attacks (HaddadPajouh et al., 2021).

2.3 IIoT Architecture

IIoT plays a crucial role in the Industry 4.0 revolution. This technology enables machine-to-machine communication and a series of automated functions within the Industrial sector. Khujamatov et al. (2021) state that in general architectural terms, IoT and IIoT are similar. However, IIoT presents an industrial-oriented architecture.

Bibi et al. (2021) reaffirms the similarity by highlighting that MQTT is a secure and lightweight protocol used for collecting data from edge devices through a publish/subscribe communication model, and are able to connect multiple devices without overloading the network. Edge devices are also utilized as gateways, acting as intermediaries between downstream devices and the IoT hub. In this research, the author brings the concept of supervisory control and data acquisition (SCADA), which is a system of hardware and software elements unique to IIoT domain.

Similarly, Boyes et al. (2018) highlights that core SCADA operations include collecting real-time data or information, and continuously monitoring the critical infrastructure, providing a consistent connection between local and remote servers. The author also lists 3 major

components that are also particular to IIoT: Intelligent Electronic Devices (IEDs), Substation Controllers, and Power Equipment. Communication occurs between substation controllers and IEDs, between IEDs and sensors, and between different IEDs inside of the same SCADA network. However, SCADA networks that were typically isolated and highly protected are now often connected to other corporate networks or the cloud, therefore they no longer hold the private or segmented status. In terms of security, this is a hack surface for external threats that can reach SCADA networks.

Recent studies also highlight that scalability, interoperability across heterogeneous devices, security, privacy, reliability, and low latency are requirements of an IIoT architecture. In contrast to the architectural models mentioned in this current study, Mirani et al. (2022) lists three main IIoT references: RAMI4.0, IIRA, and Open Fog RA. In this context, emerging technologies solutions such as edge and fog computing, blockchain, software-defined networking (SDN), 5G communication, machine learning techniques, and wireless sensor networks are being incorporated into traditional IIoTs. These new technological mechanisms are seen as solutions to reduce latency, improve interoperability, strengthen security and enable large-scale data processing across heterogeneous IIoT networks.

In addition to these protocols, Bernal and Sridhar (2023) lists, among a complete series of IIoT architectural guidelines, PROFINET and Open Platform Communication Unified Architecture (OPC UA). PROFINET is a communication protocol in the Network Layer, that relies Ethernet to connect machines, controllers, and devices to provide fast, reliable and continuous data exchange. PROFINET applies producer/consumer communication so both controller and device, that can be a sensor or actuator, sends data independently. In terms of speed, it suppresses the Request-Response model, enabling constant, real-time data exchange, which is a fundamental role in industrial automation. However, IIoTs still operate without standardization, leading to interoperability issues, strain in connecting different machines and protocol conversion demand. The OPC UA emerges as a solution to these problems, offering a standardized data model for over 60 industrial equipment with robust security mechanisms that include encryption, authentication, authorization and X.509 certificate-based key exchange. OPC UA protocol enables communication between different machines by using the same structure to describe data. The author also highlights that their applications are complementary to each other, since PROFINET is related to real-time communication solutions, and OPC UA relates to the standardization of secure data exchange across different devices and protocols once interoperable.

Several architectures have been explored to address remaining limitations of IIoT systems. Mirani et al. (2022) deliberate on the dependency of IIoTs on the deployed systems and the set of communication protocols. The research iterates that the lack of testing and the usage of new protocols compromise efficiency and reliability of IIoT networks. IIoT devices are continuously producing large amounts of data and traditional IIoT systems, which can cause transmission delays and computational offloading problems. Therefore, Mirani et al. (2022)

proposes Software-Defined Networking (SDN) to improve network management by enabling priority-based transmission control. SDN is a way of separating network components that are decision makers from those that are only able to forward packets, such as physical devices. Therefore, with the implementation of SDN controllers, the IIoT network has a central controller that has a comprehensive view of the network that allows it to define routing rules and adjust priorities, paths, and resource allocation, and contribute for efficiency enhancement. In this context, SDN usage becomes a good practice in IIoT architecture strategy that enables the system to handle high data volume, reducing latency, and supporting more adaptable and scalable IIoT networks.

Despite all the architectural advances, the IoT and IIoT networks are continuously growing and connecting more devices, expanding the attack surface. As highlighted by (Sinha, 2025), device heterogeneity, weak protocols, and communication patterns contribute to creating vulnerabilities to be exploited by attackers. Features such as wireless connectivity, cloud services, and edge processing boost the systems, but on the other hand, they create more entrance points for IoT and IIoT environments. The result can be observed in more networks being targeted by malware, botnets, and DDoS attacks. Therefore, understanding how contemporaneous research is discussing the architecture of these domains is essential for analyzing security challenges. This leads to how literature is exploring the most common threats that affect IoT and IIoT system and their possible mitigation strategies.

2.4 Attacks Taxonomy, Detection and Mitigation

2.4.1 Recon

Attacks in IoT can take many forms; however, most attacks employ Reconnaissance (Recon) as first step. It is basically an entry door for more sophisticated attacks, such as botnets and malware injections. In this type of attack, the threat actors' objective is to gather intelligence that enables them to identify possible vulnerabilities in devices and services (Alani & Damiani, 2023).

Gvozdenovic et al. (2024) proposed the IoT-Scan tool, which performs network reconnaissance for IoT devices through software defined radio (SDR), such as Zigbee, Bluetooth, and Z-Wave. It is a multi-protocol and multi-canal tool, which means that it can detect different devices that operate under different communication protocols and different frequencies. The combination of passive and active scanning was a key point for reducing time when identifying devices and mapping the network. Fingerprint devices and identify points where devices could be vulnerable.

In recent studies, Sharmin and Kiekintveld (2023) proposed a machine-learning-based approach to predict and analyze recon attacks using flags and time-based attributes. From her work, it was possible to differentiate Recon traffic from benign IoT traffic by using specific

patterns in packet timing, TCP flag combinations, and connection attempts - as SYN flags, as representation. The proposed model shows accuracy over 90% in various Recon attacks also flexibility to be extended to the analyzes of other type (DDoS).

2.4.2 Brute Force

Brute Force attacks aim to break or decrypt codes, typically by software that generates different combinations of user login and password, and performs multiple attempts until gaining access to the IoT device. Their high occurrences rely on the fact that credentials are weak, nonexistent, or rarely updated (Otoom et al., 2023).

Stiawan et al. (2019) conducted an investigation that analyzed statistical patterns and behavior in IoT traffic and demonstrated that brute force patterns can be identified by statistical analysis of features such as a rapid sequence of failed login attempts, high packet rates and repetitive login commands. This highlights that even simple security measures, such as proper authentication mechanisms at an early stage, can prevent IoT attacks.

2.4.3 Spoofing

The general meaning of spoofing is when a node can impersonate other nodes by using its IP/MAC address. Khan et al. (2022) proposed an algorithm for detecting spoofing attacks in IoT using Received Signal Strength (RSS) and Number of Connected Neighbors (NCN). In the proposed approach, IoT network are divided into cluster that represents a group of neighbor nodes and it is maintained by the core node. The combination of techniques is justified by the fact that RSS can detect, locate, and eliminate a spoofing attack. However, when it comes to intra-cluster spoofing attacks - where infected nodes operates within the same node as legitimate devices - a second mechanism must be applied to perform the same tasks. The NS-2 simulator was used to identify spoofing attacks, and the parameters of delay and negative ack were used to determine whether an spoofing attack was occurring. The negative impact was especially observed in healthcare systems where the patient's life was at risk, since spoofing attacks affect the IoT device performance. In this context, delay and energy consumption parameters were considered indicators of spoofing activity, since increased delay combined with higher energy consumption may reduce cluster lifetime and compromise system reliability.

2.4.4 Web-Based Attacks

Web-based attacks exploit vulnerabilities at the Application Layer, where many IoT devices rely on HTTP protocols, APIs, and web-based interfaces. This layer can be understood from both the user and server side. From the server side, it integrates the Web server, Web application and Database server. On the other hand, the user's side is responsible for the interface and the way the user can interact with the system. Because these components process user inputs, device configuration requests, and command execution, they become attractive entry

points for attackers to inject malicious scripts, gain unauthorized access or manipulate commands (Alaoui & Nfaoui, 2022). Since web services operate at the Application Layer, they are vulnerable to attacks such as SQL injection, Command injection, Backdoor malware, Uploading Attack, and Cross-site Scripting (XSS) (Neto et al., 2023).

2.4.5 DoS

All IoT architecture layers are susceptible to Denial-of-Service (DoS) attacks. These attacks aim to disrupt the availability of a system by exhausting its resources, preventing legitimate entities from access IoT end-node resources (Li, 2017). Many techniques can be applied to achieve the attacker's goal, such as communication blocking and network flooding.

Kadri et al. (2024) point out that single IoT devices, entire network and service can be affected by DoS attacks. This context defines the DoS attack, as it states that the main goal of a DoS attack is not to steal data or break into systems but to prevent the device or service from working properly.

M. H. Ali et al. (2022) demonstrates different DoS attacks that operate at the Perception Layer, where IoT nodes are energy-constrained and have low processing capacity. According to the author, battery draining, sleep deprivation, and node outage attacks are powerful consequences because devices rely on small batteries that limit their ability of defending from repeated or intensive requests. These attacks expose how vulnerabilities in node size, power capacity, and physical accessibility can directly compromise IoT availability.

The same author also lists DoS attacks in the Network Layer. Unlike the previous one, DoS attacks at this layer standardly aim to flood the system, while attacks such as jamming primarily block wireless transmission. Jamming can be intermittent, where attacks occur periodically or continuously. The first active jamming still sends and receives data, however, with delays and losses. Here, the service's performance is affected, and the system becomes unreliable. However, continuous jamming could take down an entire system.

2.4.6 Mirai (Case study from 2016)

The Mirai botnet emerged late in 2016 and compromised hundreds of thousands of IoT devices that temporarily disrupted activities of major Internet providers and infrastructures (Griffioen & Doerr, 2020). The main vulnerability exploited was the weak defenses and protection on these devices. This event has marked the beginning of a new era, as a consequence of Mirai's source code being largely available, allowing new codes variants to be created (Zhang et al., 2020). Therefore, a great volume of research is available in this field. Griffioen and Doerr (2020) analyzed Mirai from an epidemiological perspective, and the findings were that specific Mirai variants were frequent in some networks and botnet developers tend to repeat the patterns of their competitors. However, the research concluded that IoT botnets are not self-sustaining and tend to reach a constant state or eventually die out. Despite the

natural lifecycle of the botnet, Mirai remains strongly associated with the distributed term, strengthening its dependency of a massive number of compromised IoT devices.

Alwaisi (2025) proposed a TinyML-based real-time anomaly detection mechanism that could efficiently classify the UDP flood, TCP flood, SYN flood, and Scan attacks – all of them are types of Mirai botnet attacks. The proposed dispositive revealed metrics of evaluation exceeding 99% when using KNN approach, followed by SVM, which presents similar precision rates of 97,8%.

Another approach for detecting Mirai was proposed by Palla and Tayeb (2021), but now in two different programming environments: Matlab and Python. By analyzing Mirai and Benign datasets that were split into training, validation, and test sets, the model trained an Artificial Neural Network (ANN) with neurons in the hidden layer that produced results of 92.8% accuracy, 0.3% False Negatives, and 0.99% F-Score.

Overall, the Mirai case highlights how large-scale botnets enable distributed attacks such as DDoS (Sahu et al., 2021).

2.4.7 DDoS

Distributed Denial of Services (DDoS) has the same goal as DoS, which is preventing a service from performing its regular function. However, the DDoS has more performance complexity, and attacks originate from distributed compromised devices.

According to De Donno et al. (2018), DDoS attacks involve execution stages such as device infection, botnet recruitment, command-and-control, and flooding attacks until services are denied to their legitimate users.

The report Khalil (2025) reinforces that the rapid growth of unprotected devices from the IoT can be directly related to the number of IoT attacks, which in 2025 accounted for 820,000 IoT attacks daily - an increase of 42% compared to the statistics from last year, mostly automated driven, trying to exploit device vulnerabilities. It also indicates that Brute Force exploits vulnerabilities in credentials, compromising the devices with protocols such as Telnet. Once these devices are compromised, they are typically incorporated in large-scale botnets that perform coordinated DDoS attacks. In the same scenario, Netscout's (n.d.) reported more than 8 million global DDoS attacks in the first half of 2025, including over 50 incidents where 1 Tbps and a record of 3.12 Tbps.

The progressive increase in poorly protected IoT devices, and the sophistication and range of DDoS attacks show that traditional static defenses are no longer effective to mitigate attacks in IoT environments. Modern IoT botnets generate highly dynamic traffic patterns, distributed across the network and so advanced in their structure that rule-based systems cannot distinguish them from legitimate activity. Therefore, Machine Learning techniques become essential, since even the most subtle anomalies can be detected compared to normal traffic,

adapting their strategy over time (Alani & Damiani, 2023). Machine learning techniques can also be applied at large scale, overcoming the limitations of human-based analysis (Neto et al., 2023).

Moreover, Singh and Jain (2025) provide a survey in IoT networks using software-defined networking, which reinforces the affirmation that IoT networks are large, dynamic and heterogeneous. In this way, traditional rule-based defenses are no longer effective. The authors also highlight that ML models trained on network traffic can distinguish, in real-time, the benign and abnormal activity and also detect attacks. The survey strengthens the debate about the indispensability of ML techniques – or even more advanced approaches - for securing IoT environments.

The use of machine learning has proven to be effective and has high performance in reducing or mitigating the security challenges in the IoT. Roopak et al. (2020) studied feature selection, choosing specific attributes with more relevance in network traffic to train the models, aiming at improving the detection of DDoS attacks in IoT networks.

In this context, the complexity development, scale, and variability of DDoS attack highlight the limitation of traditional detection mechanisms. While DDoS traffic becomes more dynamic and harder to distinguish between legitimate activity, intelligent and adaptive detection mechanisms are needed. Therefore, machine learning-based approaches emerge as an efficient detection approach, enabling the identification of malicious activity in IoT and IIoT networks (M. H. Ali et al., 2022).

2.5 Machine Learning for IDS

Machine learning techniques have been widely used in IDS to analyze network behavior patterns and identify abnormal traffic. By using data to train models, this approach detects both known and unknown attacks, reinforcing that machine learning-based approach are suitable for dynamic and heterogeneous environments such as IoT and IIoT. In this context aspects not captured by rule-based IDS methods may be captured by machine learning models (M. H. Ali et al., 2022).

Supervised Learning and anomaly based detection often represent machine learning approaches in IDS. In supervised learning, models are trained using labeled data to classify traffic into predefined categories. This is commonly used in DDoS detection, where classification models learn from these labeled data and further are used to distinguish between benign and DDoS traffic. On the other hand, anomaly-based techniques identify deviations from legitimate traffic spotting unusual or potential malicious activity. Therefore, both approaches rely on feature behavior analysis, making them effective tools for detecting abnormal traffic in environments such as IoT and IIoT (Kayode Saheed et al., 2022). Therefore, it is essential to understand how features represent network behavior. For instance, features such as ports

and protocols are widely represented by numbers. However, they are not solely numerical values, as they reflect other aspects of functional communication, such as which service is being accessed (Duc Nguyen et al., 2018).

Models such as Random Forest (RF) and Support Vector Machines (SVM) are types of supervised learning approaches are frequently used in intrusion detection due to their effectiveness in dealing with network traffic data. Random Forest builds multiple decision trees by combining their outputs to improve classification performance. This is particularly effective for complex network scenarios where numerical features can be both, continuous and categorical as highlight Ebady Manaa et al. (2024). Differently, SVM are used by identifying decision boundaries that separate between different classes, while maximizing the margin between them, making this type of classification model suitable for distinguishing between traffics. In addition, they have been applied in IoT environment, demonstrating a reliable tool for handling high-dimensional data.

However, despite the relevance of machine learning techniques in IDS, the literature has also identified significant limitations. According to Jullian et al. (2023), traditional approaches rely on the quality and representation of features, making their selection a critical aspect of reliable performance. In addition, small variations of existing attacks can negatively impact the detection accuracy in highly dynamic environments. As a result, different domains or datasets may present different model performance. These limitations directly align with the importance of understanding the behavior of features, rather than simply automating this process.

Although the same foundational networking technologies are shared between both domains, they are unique in other characteristics, such as scalability, reliability, and real-time communication requirements (Leng et al., 2022). The heterogeneity of IoT environments, including their devices, protocols and communication patterns may also influence traffic behavior (Casola et al., 2019). Therefore, models can have a different performance depending on the environment which they are applied, making feature selection and scalability a remaining challenge in IoT and IIoT security studies. Therefore, understanding feature behavior is a key factor when analyzing intrusion detection models across domains.

2.6 Research Gaps

Although literature has shown significant progress in developing mechanisms for detecting anomalies, especially those with machine learning-based approaches, several methodological gaps remain. Firstly, most literature relies their studies on a single dataset, which represents a limitation for cross-domain analysis. However, feature selection and model performance end up becoming useful to that specific dataset only, resulting in inefficient solutions when applied to different datasets that contain an unsimilar feature set or present data collected in a different domain, which has different protocols and device behaviors. The under-

lying problem is that feature behavior are underexplored under different conditions; therefore their stability, relevance, and redundancy cannot be understood out of a given context.

Secondly, the literature also shows insufficient evidence-based analysis of feature behavior across time. Considering that IoT and IIoT devices are rapidly growing without a standardized pattern, the combination of this heterogeneous network with attackers' ability to find new exploitation strategies reinforces that models trained in a single dataset or collected in a specific time are at risk of becoming outdated. As highlighted, botnets such as Mirai have an epidemiological characteristic that corroborates demand for cross-year analysis (Griffioen & Doerr, 2020).

Despite the architecture differences, especially related to how industry-specific IIoT is, literature often treats both domains homogeneously. The same was observed in statistical analysis, which frequently includes IIoT metrics in IoT without properly highlighting their differences. Overall, the absence of domain-oriented research results in under-examination of technological and behavioral distinctions. For instance, the IoT applies to lightweight protocols, such as MQTT, CoAP, BLE, and Zigbee (Choudhary, 2024). IIoT architecture, however, includes industrial protocols such as PROFINET, EtherCAT, OPC UA, Modbus (Bali et al., 2021; Trifonov & Heffernan, 2023). These differentiations are fundamental when analyzing traffic patterns, resulting in different anomaly manifestations for each domain.

This highlights the need of cross-domain analysis focusing on feature behavior, aiming to understand how different domains and datasets influence on feature relevance.

3

Methodology

This chapter lists the research methodologies applied to achieve each one of the objectives of the present study. Therefore, the methodological process must ensure that at the end of the project an evidence-based guidelines for feature selection in IoT and IIoT DDoS detection will be developed. The following sections discuss each of the study's research methods in turn.

3.1 General methodology overview

This project adopts a methodological research approach that includes exploratory, descriptive, and experimental techniques. The general purpose of this methodological strategy is to examine feature behavior in IoT and IIoT network traffic, by statistically assessing these patterns and validating the findings by performing results experimentation.

Accordingly, the methodology follows a sequential structure, where the process is initiated with the literature review, a qualitative research methodology to gather information on how past research has explored the subject and what topics are yet underexplored or not fully understood. This step is followed by the selection of the datasets utilized further in the experimentation phase, which is described in a dedicated section (Pilcher & Cortazzi, 2024).

Descriptive research is performed to explore the dataset through an Exploratory Data Analysis (EDA) that describes how features behave, enabling the study to proceed with a quantitative statistical approach. In this phase, features are statistically explored using methods to identify their relevance, redundancy, and domain dependence.

Results from previous stages are validated through quantitative experimentation analysis according to the explanation in the 3.4 section. Together, these methods provide a coordinated approach for evaluating characteristic behavior and supporting the development of practical evidence-based guidelines.

3.2 Research Method: Literature Review

As a starting point, the literature review is the first research method applied to the current project, aiming to address both SO1 and SO2, which refers to comparative feature behavior analysis, considering unique characteristics of each domain and identifying feature relevance, redundancy, and domain dependence. In this context, this section identifies the key concepts of IoT and IIoT networks, along with their respective architectures, challenges, and research gaps. In other words, this *qualitative* research methodology is the first step that enables the development of theories by gathering studies from different perspectives, enabling the identification of patterns and divergences across the literature.

Furthermore, it guides us through a meta-level understanding of a specific subject and highlights areas where their research is still dubious, cloudy, under-covered, or inexistent (Snyder, 2019).

This literature review is conducted by a systematic search in major academic databases, including, but not limited to, journal articles, conference proceedings, and reports. Primarily, the search is conducted using ScienceDirect, Springer, IEEE Xplore, and ACM Digital Library applying search terms such as: “Internet of Things”, “IoT”, “Industrial Internet of Things”, “IoT Architecture”, “IIoT Architecture”, “IoT vs. IIoT”, “IoT security challenges”, and “IoT Machine Learning”. In parallel, Google Scholar was also used for gathering academic research as the standard Google search tool was used for less technical information retrieval, such as economic impact and applications.

The selection prioritized recent publications and sources recognized by their credibility and academic relevance, especially in the technological field.

3.3 Research Method: Data Analysis

This section describes the data analysis methodology used in this research, including dataset selection, exploratory data analysis, categorical feature examination, correlation analysis, identification of protocol-dependent features, discriminative power analyzes through RBC, and feature-level analysis with the aim of creating reduced datasets.

3.3.1 Dataset Selection

In order to support the research objectives of this study, datasets are required to enable the analysis of feature behavior across different domains, as well as the validation of their dis-

criminative power in DDoS detection through supervised machine learning models. Therefore, the selected datasets must provide information that enables feature behavior analysis in IoT and IIoT environments, including labeled benign and DDoS traffic.

Existing literature reflects the vast availability of datasets for IoT and IIoT research. However, they differ in terms of structure, extraction, feature representation, and attack scenarios. Addressing the comparative and cross-domain analysis from SO1 and the identification of relevant features for DDoS detection, two datasets were selected: CIC IoT-DIAD 2024 and CIC IIoT 2025.

Together they provide representation of the network environments covered by this study and complementary data representation, as they are flow-based and time-window-based. This enables feature behavior analysis under different conditions.

IoT Dataset – CIC IoT-DIAD 2024

The CIC IoT-DIAD 2024 dataset (Rabbani et al., 2025) was developed by the Canadian Institute for Cybersecurity (CIC) collecting key information to support both IoT device identification and anomaly detection. It contains data of 33 different attacks performed in 105 different devices.

The dataset was produced using CICFlowMeter open tool, in which pcap files are processed to produce biflows, enabling flow-level features extraction. The dataset is also organized into TCP and UDP flows, in which TCP flows terminated upon connection teardown and UDP flows terminated based on a predefined timeout.

The dataset contains 8 distinct folders divided by traffic type, including 1 benign and 7 attack categories. However, aligned to the scope of this project, only benign and DDoS traffic are considered for the analysis.

IIoT Dataset – CIC IIoT 2025

The CIC IIoT dataset 2025 (Firouzi et al., 2025) was developed by the Canadian Institute for Cybersecurity to provide a realistic benchmark for attack analysis in industrial environments. To address the heterogeneity of devices and communication protocols gap, the Institute dedicated a IoT and IIoT laboratory with 40 interconnected devices, including 15 industrial sensors.

These devices allowed traffic and sensor data to be collected in a synchronized way over fixed time windows ranging from 1 to 10 seconds, representing the timeframe of IIoT activity. This results in a dataset including benign and multiple attacks scenarios providing time-window-based feature representations.

Structural Differences

Both datasets were selected because they support comparative analysis across different domains, aligning with SO1. Although they were developed by the same institution, their different structures directly impact the analysis.

3.3.2 Exploratory Data Analysis

Addressing SO1 of the current work, this study performs an extensive Exploratory Data Analysis (EDA) for each domain individually. This descriptive research approach aims to understand feature behavior within each dataset by analyzing how traffic patterns and dataset characteristics influence feature distribution.

To support the EDA, features are analyzed at a global level to understand the statistical behavior of each dataset. At this stage, measures of central tendency (mean and median) are compared to identify whether features are asymmetric, and measures of dispersion (standard deviation and variance) are used to assess the level of variability within each domain. Furthermore, features presenting higher variability are further analyzed, as they may reflect important fluctuations in network traffic behavior (Z. Ali et al., 2019). This investigation supports the identification of patterns and abnormal traffic that guide further protocol-dependency and feature level analysis.

3.3.3 Categorical Feature Analysis

Features such as port are treated as categorical representations in this study and are not limited to their numerical values. Therefore, a specific analysis focusing on destination ports is conducted, since they represent services being accessed. This is justified by the fact that destination ports provide a more consistent interpretation of pattern behavior, while source ports are dynamically assigned and primarily used to identify individual connections, particularly in mechanisms such as Port Address Translation (PAT) (Duc Nguyen et al., 2018).

Destination ports are then examined to understand the frequency at which they are used and also the volume of packets transmitted through them. Since this information was made available differently in each dataset, the analysis was adapted accordingly. In the IoT dataset, it is possible to verify with precision both the frequency and the volume associated with each destination port. However, in the IIoT dataset, each observation may contain multiple destination ports, limiting the precise determination of the volume of packets transmitted for each destination port. To address this, the data is restructured to enable the individual examination of ports; however, the analysis is limited to evaluating the volume in relation to the number of ports identified within each observation. This adaptation enables that destination port behavior can be analyzed consistently across both datasets despite the datasets structures.

3.3.4 Correlation Analysis

The correlation analysis between features is carried out after the data from the EDA is organized and structured, allowing the identification of potential redundant features and how strongly they are related to other features within the same domain. This approach is justified by the fact that some collected data capture the same information, but in different ways. Therefore, high correlation may indicate redundant features. In this sense, the correlation coefficient (r) can be used to identify redundancy between features, directly addressing SO2.

The Pearson correlation coefficient (r) is used to quantify the strength and direction of the linear relationship between variables for each dataset. Features with high positive correlation are evaluated for their potential redundancy, as indicates that these features capture the same information.

The analysis is conducted individually for each domain. In addition, correlation heatmaps are used to visualize feature relationships, supporting the identification of highly correlated variables. Further on, these features are examined to identify whether they are relevant for understanding traffic pattern behavior.

In this sense, identifying redundant features is an important step to achieve SO3, since performing experimental validation through testing machine learning models also involves reducing data noise for better classification model performance (Jo et al., 2019).

3.3.5 Protocol and Group Feature Analysis

This subsection describes how features behave considering protocol dependency and discriminative power across both IoT and IloT environments. The IoT dataset provides specific indication of TCP (6) and UDP (17) through the protocol feature, which is used as the basis for protocol-level analysis.

Protocol dependency analysis is conducted for the IoT data set by comparing the statistical behavior of features present in both protocols, including comparison based on descriptive statistics, where the absolute difference in median and standard deviation is calculated for each feature across protocols. Features with higher variability between protocols are ranked and identified as the most influenced by the underlying protocol. Finally, features with the largest difference in standard deviation are identified as the most dependent on a specific protocol.

In contrast, the IloT dataset does not provide direct protocol labeling, as traffic is represented through aggregated features derived from different aspects of network communication, such as packet structure, timing and header information. Therefore, instead of analyzing protocol dependency, in IloT the analysis focuses on identifying which feature groups contribute most to distinguishing between benign and DDoS traffic. Features are grouped according to the

indication provided by the dataset description, which reflects their functional characteristics. This process represents a higher-level interpretation of behavior across feature categories, and precedes a more detailed feature-level analysis presented later.

To quantify the discriminative power of these groups, the Rank-Biserial Correlation (RBC), derived from the Mann–Whitney U test, is applied. The RBC enables measuring the strength of separation between two groups; in this analysis, it measures the separation between traffic classes. At this stage, the analysis is performed at the group level rather than at the individual feature level. For this, the average RBC and its variability are used to evaluate whether the features within a group consistently contribute to traffic classification, and not only isolated features within the group.

This allows the identification of feature categories that present both strong and stable discriminative behavior in the IIoT environment.

3.3.6 Discriminative Power Analysis (RBC)

In this study, the Mann–Whitney U test is applied using a two-sided hypothesis to evaluate differences between traffic classes. Here, benign and DDoS traffic are compared to determine their distribution differences, allowing the analysis to capture any type of variation between the classes (Salvatore S. Mangiafico, 2026).

While for IoT the discriminative power is calculated at a feature-level using RBC, for IIoT this step is used to refine the group-level analysis performed previously, narrowing down the identification of the most relevant features.

3.3.7 Feature-Level Analysis and Dataset Reduction

Following the same discriminative power analysis, the study now performs a feature-level evaluation to identify the most relevant attributes for distinguishing between traffic classes. While the evaluations so far rely on statistical methods, such as Pearson Correlation and RBC, this stage combines statistical results with insights from the existing literature to support feature selection decisions.

The combination of these methods aims to identify which features are redundant, through correlation, and which features have higher discriminative power according to the RBC. However, since this analysis is initially performed individually for each dataset, additional information from the literature is considered to support a more generalizable feature selection process across IoT and IIoT environments.

For this, features are analyzed both individually and within their groups, and those with higher RBC values are considered more relevant, as they better distinguish between traffic classes. However, discriminative power alone is not sufficient for feature selection. When features from the same group share a strong correlation ($r > 0.95$), they are considered redundant.

In that case, only the most representative feature is retained to reduce redundancy and avoid overlapping information (Jo et al., 2019).

However, as the selection considers multiple factors, features with slightly lower RBC values may still be retained as complementary attributes when they capture different aspects of the same behavior, such as variability across traffic instead of magnitude.

Conversely, highly correlated groups combined to low RBC are excluded from further analysis. The features excluded are mostly associated with near-constant values, weak or absent statistical differences between traffic classes, or their behavior were already captured by other feature.

This step results in the creation of a reduced dataset for each domain with features that preserve the most relevant characteristics of network traffic while reducing noise and redundancy. The new datasets are then used in the following stage of the study, where machine learning models are applied to evaluate whether the features selected captures relevant information for DDoS detection.

3.4 Research Method: Experimentation

Satisfying SO3, at this phase, this research adopts an experimental methodology that aims to verify whether the findings obtained from statistical inferences, combined with insights from the literature, are aligned with the proposed feature selection guidelines. Therefore, two baseline machine learning models are used, Random Forest and Support Vector Machine (SVM), to confirm whether the features identified in previous stages improve consistency when detecting DDoS traffic across both explored environments (Géron, 2018).

The reduced feature set obtained from the previous stage is used to train these models, and their performance is evaluated considering their ability to differentiate between traffic classes. This approach aims to identify whether the selected features contribute to DDoS detection.

3.4.1 Evaluation

Each model performance is evaluated using standard classification metrics, such as accuracy, precision, recall, and F1-score, which measure different aspects of model performance, including correctness, positive cases identification, and the balance between precision and recall. In this sense, multiple metrics are considered, as they capture different aspects of the classification results and may provide better insights, especially in scenarios where class imbalance affects the interpretation. The evaluation supports SO3 by experimentally validating the features previously selected.. (Luque et al., 2019).

3.4.2 Cross-domain analysis

Together, the results provide a solid and comprehensive contextual basis for performing a cross-domain analysis, where it is possible to identify which features have consistent behavior patterns across environments and which behave differently.

3.5 Evidence-based guidelines for feature selection in IoT-IIoT networks

The final step and artefact of this research consists in consolidating the evidence collected from the EDA, statistical analysis feature selection based on statistical and analytical criteria and experimental validation, aiming to develop practical guidelines for identifying and selecting relevant features for DDoS detection, directly addressing the primary objective of this research. The guidelines are fruit of a structured process of statistical interpretation, results comparison, and argumentation, where findings from different stages are combined to support feature selection.

The guidelines identify features by their category rather than their representation in a single dataset, allowing the artefact to be useful for future research adopting a machine learning approach, by providing an evidence-based reference for best practices in feature selection.

4

Implementation and Results

This chapter presents the implementation and the results obtained from the application of the methodology described in Chapter 3. The implementation details the application of the proposed framework, including data preparation, statistical analysis, feature selection, and model validation for both IoT and IIoT datasets.

The process starts with EDA performed individually for each dataset to understand the general feature behavior, including feature distribution and the importance of destination ports. This is followed by correlation analysis and the identification of protocol-dependent features. However, due to the nature of the dataset, in the IIoT the analysis focuses on the discriminative power of feature groups.

Additionally, following the methodological framework, a feature-level analysis is conducted, combining statistical evidence from the datasets with insights from the existing literature. The selected features are further evaluated through experimental validation. Baseline machine learning models described in Chapter 3 are used to verify whether the statistically identified attributes effectively improve DDoS detection across IoT and IIoT environments.

Furthermore, the results support the development of evidence-based feature selection guidelines.

4.1 Implementation

This section presents the implementation of the methodology described in Chapter 3. Describes how data are prepared, statistical analysis is performed, features are selected, and how models are used in the validation process.

Due to the different structure present in the IoT and IIoT datasets, adaptations in the implementation are required to maintain the same level of analysis. The goal is to maintain consistently the results and to provide support for creating reduced datasets used in the machine learning models.

4.1.1 IoT Data Preparation

The implementation is conducted exclusively using Python for data manipulation and pre-processing. The first part consists in constructing the IoT dataset by merging multiple CSV files into a single dataset. During this process the benign and DDoS traffic are identified and labeled, as support for further analysis.

The data cleaning is applied to ensure result consistency. Therefore, infinite values are replaced with missing values. Infinite values were only encountered in Flow Bytes/s, Flow Packets/s. These features are derived from division by time-related variables, which may result in infinite number values when the duration observed is zero.

Only numerical features are retained for statistical analysis, as the focus of this project is statistical behavior analysis between variables. The original joined dataset consists of 86 features. However, after selecting only numerical ones, only 79 features remained, as shown in Table

Table 4.1: Numerical features used in the CIC IoT-DIAD 2024 dataset

Feature	Feature	Feature
Flow duration	total Fwd Packet	total Bwd packets
total Length of Fwd Packet	total Length of Bwd Packet	Fwd Packet Length Min
Fwd Packet Length Max	Fwd Packet Length Mean	Fwd Packet Length Std
Bwd Packet Length Min	Bwd Packet Length Max	Bwd Packet Length Mean
Bwd Packet Length Std	Flow Bytes/s	Flow Packets/s
Flow IAT Mean	Flow IAT Std	Flow IAT Max
Flow IAT Min	Fwd IAT Min	Fwd IAT Max
Fwd IAT Mean	Fwd IAT Std	Fwd IAT Total
Bwd IAT Min	Bwd IAT Max	Bwd IAT Mean
Bwd IAT Std	Bwd IAT Total	Fwd PSH flags
Bwd PSH Flags	Fwd URG Flags	Bwd URG Flags
Fwd Header Length	Bwd Header Length	FWD Packets/s

Feature	Feature	Feature
Bwd Packets/s	Packet Length Min	Packet Length Max
Packet Length Mean	Packet Length Std	Packet Length Variance
FIN Flag Count	SYN Flag Count	RST Flag Count
PSH Flag Count	ACK Flag Count	URG Flag Count
CWR Flag Count	ECE Flag Count	down/Up Ratio
Average Packet Size	Fwd Segment Size Avg	Bwd Segment Size Avg
Fwd Bytes/Bulk Avg	Fwd Packet/Bulk Avg	Fwd Bulk Rate Avg
Bwd Bytes/Bulk Avg	Bwd Packet/Bulk Avg	Bwd Bulk Rate Avg
Subflow Fwd Packets	Subflow Fwd Bytes	Subflow Bwd Packets
Subflow Bwd Bytes	Fwd Init Win bytes	Bwd Init Win bytes
Fwd Act Data Pkts	Fwd Seg Size Min	Active Min
Active Mean	Active Max	Active Std
Idle Min	Idle Mean	Idle Max
Idle Std		

To support the protocol-level analysis, it is required to filter the protocol features to select only TCP (protocol 6) and UDP (protocol 17). Then, they are separated into subsets enabling independent feature behavior evaluation for each protocol.

New variables are created conform needed to support the analysis. Particularly in the port analysis the variable `Total Packets` is created by joining `Fwd Packets/s` and `Bwd Packets/s`, enabling for a more informative representation of traffic behavior.

Finally, the dataset is organized to support further stages of the implementation, including statistical analysis, correlation, protocol-dependency and feature-level evaluation. These steps ensure consistency for evaluation procedures applied in the next stages.

4.1.2 IIoT Data Preparation

The implementation follows the same preprocessing pipeline described for the IoT dataset, in which Python is used for data manipulation. The IIoT dataset is created by merging two distinct CSV files containing benign and attack classification, collected over fixed time windows of 1 second. During this process, only DDoS samples are selected from the attack dataset, creating binary target variables, in which benign and DDoS are represented for 0 and 1 respectively. At the first implementation stage, the dataset consists of 227,191 samples, comprising 136,800 benign and 90,391 DDoS observations.

Additionally, the original dataset contains multi label representation, including attack type and categories that are necessary for the following analysis, and therefore removed.

When merging the datasets, features stored as lists such as IP addresses, ports, protocols, and MAC addresses are converted into strings and converted back to the original format

using literal evaluation, preserving their structural representation. No missing values are identified on the 90 features. However, only 72 features with numerical representation are retained for statistical analysis, as shown in 4.2. Non-numerical features including identifiers, timestamps, and list-based attributes that are not statistically compatible with the statistical methods applied in this study.

The dataset is finally organized to support further stages of the implementation. However, due to its aggregated and structured nature, some adaptations are required in later stages.

Table 4.2: Numerical features used in the CIC IoT 2025 dataset

Feature	Feature
log_data-ranges_avg	log_data-ranges_max
log_data-ranges_min	log_data-ranges_std_deviation
log_data-types_count	log_interval-messages
log_messages_count	network_fragmentation-score
network_fragmented-packets	network_header-length_avg
network_header-length_max	network_header-length_min
network_header-length_std_deviation	network_interval-packets
network_ip-flags_avg	network_ip-flags_max
network_ip-flags_min	network_ip-flags_std_deviation
network_ip-length_avg	network_ip-length_max
network_ip-length_min	network_ip-length_std_deviation
network_ips_all_count	network_ips_dst_count
network_ips_src_count	network_macs_all_count
network_macs_dst_count	network_macs_src_count
network_mss_avg	network_mss_max
network_mss_min	network_mss_std_deviation
network_packet-size_avg	network_packet-size_max
network_packet-size_min	network_packet-size_std_deviation
network_packets_all_count	network_packets_dst_count
network_packets_src_count	network_payload-length_avg
network_payload-length_max	network_payload-length_min
network_payload-length_std_deviation	network_ports_all_count
network_ports_dst_count	network_ports_src_count
network_protocols_all_count	network_protocols_dst_count
network_protocols_src_count	network_tcp-flags-ack_count
network_tcp-flags-fin_count	network_tcp-flags-psh_count
network_tcp-flags-rst_count	network_tcp-flags-syn_count
network_tcp-flags-urg_count	network_tcp-flags_avg
network_tcp-flags_max	network_tcp-flags_min

Feature	Feature
network_tcp-flags_std_deviation	network_time-delta_avg
network_time-delta_max	network_time-delta_min
network_time-delta_std_deviation	network_ttl_avg
network_ttl_max	network_ttl_min
network_ttl_std_deviation	network_window-size_avg
network_window-size_max	network_window-size_min
network_window-size_std_deviation	target

4.1.3 Statistical Analysis Execution

This stage describes the statistical analysis applied for both datasets. It flows as the sequential approach starting with descriptive statistics, extending to global correlation, protocol-dependency evaluation (for IoT) and group-level evaluation (for IIoT) and feature-level discriminative analysis.

Each dataset is evaluated separately computing their global descriptive statistics, including mean, median, standard deviation, and variance.

Next, correlation analysis is performed using the Pearson correlation coefficient to evaluate relationships between features. At this stage, correlation matrices are generated for each dataset, with heatmap supporting for visualization of highly correlated features and potential redundancy.

The protocol-dependency analysis consists in separating TCP and UDP traffic from the IoT dataset. Features that share both protocols are then compared based on their statistical behavior, allowing the identification of behavior patterns when protocol changes, in other words, protocol-dependency behavior.

Due to the datasets structure, for the IIoT dataset the analysis is adapted to a group-level evaluation. The RBC is used to compute discriminative power. Sequentially, features are mapped according to their structural categories, as indicated in the dataset description classification. The results compute the average RBC calculated for each group, enabling the identification of groups that consistently differentiate between traffic classes.

Finally, the discriminative power evaluated at a feature-level in both environments using RBC, derived from the Mann–Whitney U test. This allows features to be ranked according to their capacity of distinguishing between benign and DDoS traffic. In the IIoT dataset, this analysis is first performed at the group level and later refined at the feature level.

4.1.4 Feature Selection and Dataset Reduction

The process used for feature selection and data reduction aims to identify most relevant features for describing traffic patterns, distinguishing between traffic classes while reducing

redundancy and noise.

Feature selection combines statistical evidence and structural analysis. First, RBC evaluates features that are considered more relevant for traffic differentiation. In addition, feature correlations identify potential redundant features, which are candidates for exclusion, due to noise and dimensionality reduction.

Furthermore, related features are evaluated together, such as features represented in their statistical variations (feature min, mean, std and max) or with their respective groups. This allows the evaluation of complementary features that capture different aspects of traffic behavior, even when the RBC returns a limited discriminative power.

The last part of this multifactorial feature-selection examination consists of insights from the existing literature. This is a fundamental step for reducing the dependence on a single dataset, as perspectives from previous studies enhance the generalization of these guidelines.

As a result, a reduced dataset is constructed for IoT and IIoT, preserving most representative features by minimizing overlap information. These reduced datasets are then used in the experimental validation stage with machine learning models.

4.1.5 Selected Features

Following the feature selection and data set reduction process, two reduced sets of features were defined for both IoT and IIoT that will be used in the machine learning models.

IoT

The features for the IoT dataset identified in the feature selection process are described in Table 4.3.

Table 4.3: Selected features for the IoT dataset

Feature	Feature
Total Bwd packets	Total Length of Bwd Packet
Fwd Packet Length Min	Bwd Packet Length Max
Flow Packets/s	Flow IAT Std
Flow IAT Min	Fwd IAT Std
Fwd Packets/s	Bwd Packets/s
Packet Length Max	Packet Length Mean
RST Flag Count	Average Packet Size
Bwd Segment Size Avg	Subflow Fwd Packets
Subflow Fwd Bytes	Fwd Seg Size Min
Idle Min	

IIoT

The features for the IIoT dataset identified in the feature selection process are described in Table 4.4.

Table 4.4: IIoT Reduced Dataset

Feature	Feature
log_data-ranges_max	log_data-types_count
log_interval-messages	log_messages_count
network_fragmented-packets	network_header-length_avg
network_interval-packets	network_ip-length_avg
network_ip-length_max	network_ips_all_count
network_packet-size_avg	network_packet-size_min
network_packets_all_count	network_packets_src_count
network_ports_all_count	network_ports_dst_count
network_protocols_dst_count	network_tcp-flags-ack_count
network_tcp-flags-psh_count	network_tcp-flags-rst_count
network_tcp-flags-syn_count	network_time-delta_min
network_time-delta_std_deviation	network_ttl_max
network_ttl_min	

4.1.6 Baseline Model Implementation

This stage describes the implementation of the baseline machine learning models used for experimental validation. To achieve SO3, the findings are validated in practice using two baseline models: Random Forest and Support Vector Machine (SVM), both applied to the IoT and IIoT datasets.

Each reduced dataset produced is used as input. The target variables correspond to traffic class (benign and DDoS). Although at this point the dataset contains only numerical features, this is reinforced at the model implementation.

The dataset is split into training and testing sets using an 80/20 ratio, ensuring similar class distribution across sets.

For the Random Forest model there is a fixed configuration including 50 estimators, a maximum depth of 10, and a maximum of 1000 leaf nodes. Next, the model is trained with the training set and used to generate predictions on the test set.

Similar pipeline is used for both reduced datasets to ensure fair comparison.

For the SVM model, a pipeline is implemented to ensure consistent preprocessing before classification. This pipeline includes median imputation, feature standardization, and a LinearSVC classifier that is applied for both datasets, with adaptation of class weighting added in the IIoT dataset due to class imbalance. The adaptation supports for consistent results across datasets.

4.1.7 Evaluation Procedure

Predictions are generated and compared with the true labels (y_{test}). The evaluation is based on standard classification metrics, including accuracy, precision, recall, and F1-score, allowing different aspects of the model performance to be assessed.

The same evaluation process is applied to both models to ensure comparison consistency.

The previous sections describe the implementation process to achieve the following results.

4.2 Statistical and Analytical Results

4.2.1 IoT

The statistical feature behavior for the IoT dataset is characterized by high variability, strong right-skewness, and the presence of extreme values.

When sorted and listed by standard deviation, all the top-15 features displayed were time related, such as flow duration, inter-arrival times, and idle periods. The results indicate that time-based features are strong indicators of variability in IoT traffic, as shown in 4.3

Additionally, mean values are consistently higher than the median across these features, revealing that the distribution is highly asymmetric and dominated by a small number of expressive observations, while the majority of the observations remain at low values. The dispersion measures also showed values similar or higher than the mean, with variance reaching extremely high magnitudes, reinforcing the feature's heavy-tailed distribution.

The features Bwd IAT Total, Flow IAT Std, Bwd IAT Max, and Fwd IAT Std presented median values equal to zero, indicating that at least half of the flows produce no measurable backward activity or time variability within the aggregation window. The 4.2 provides a comparison between the mean and median values of the top-10 most variable IoT features.

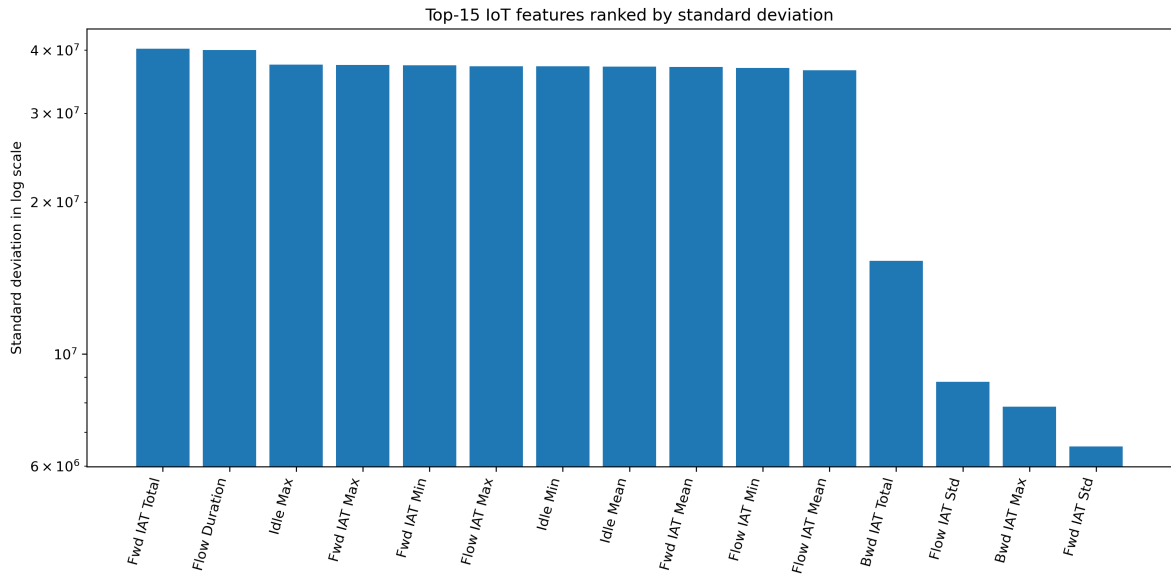


Figure 4.1: Top-15 IoT features ranked by standard deviation

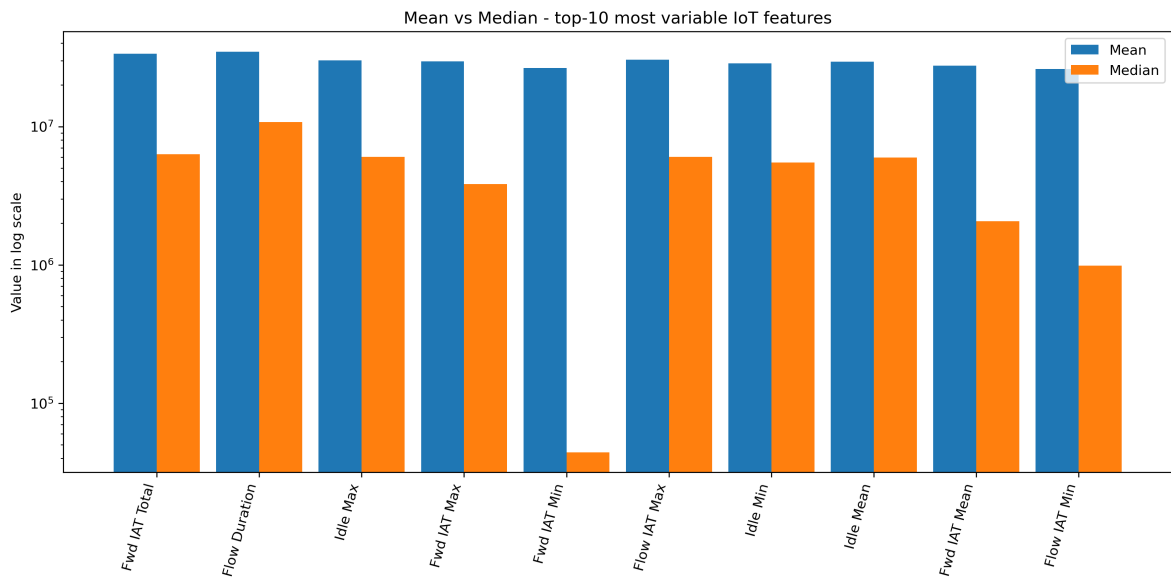


Figure 4.2: Mean vs Median - top-10 most variable IoT features

4.2.2 Industrial IoT

The IIoT dataset exhibited strongly skewed distribution patterns. Almost all features presented medians equal to zero. The same pattern was also observed among the top-15 features ranked by standard deviation. The discrepancy between measures of central tendency and dispersion, reveals that most observations exhibit low activity levels while a limited number of cases account for substantially higher values.

The highest variability was observed in features related to Packet Traffic Rate, Timing Control, Header Flags, and Network Multiplexing. In contrast, features related to Size/Length,

Address Diversity presented low variability.

4.3 Destination Port

4.3.1 IoT

Results showed a high concentration of destination ports in a small set of services, with higher frequency observed in IRC (Internet Relay Chat) port 6668, followed by DNS (53), RTSP (554), HTTPS (443), and HTTP (8080 and 80). Overall, the results suggest that IoT traffic is dominated by a limited set of destination ports.

When separating destination ports by traffic class, benign traffic presented the same destination ports identified in the global frequency analysis. However, the high frequency observed for ports 32100, 6667, and 8886 reveals greater diversity in the distribution of services. In contrast, DDoS traffic reveals a higher frequency concentration for ports listed before, while other additional ports are substantially less frequent.

The volume-based analysis further reinforces the different behaviors by showing traffic intensity distribution across destination ports. Destination ports common to both traffic classes consistently showed a higher volume in DDoS flows. Also, of the top 10 destination ports displayed, at least 2 (10008 and 20443) appeared exclusive in attack traffic. These results indicate that not only attack traffic target same ports, but also concentrates higher traffic intensity on them.

4.3.2 Industrial IoT

The destination port 1883 (MQTT) ranked as the top in both benign and DDoS traffic. Additionally, 80 (HTTP) and 8886 presented elevated aggregated packet count in DDoS traffic and port 9200 was also identified for both, benign and DDoS traffic.

The frequency-based analysis showed more diverse distribution of destination ports, while DDoS traffic concentrated attacks in ports 1883 (MQTT) and 80 (HTTP). This indicated that while benign traffic involve a broader set of endpoints, DDoS traffic is focused on specific targets.

Overall, benign traffic has a heterogeneous distribution of destination ports with similar packet count associated to these ports. In contrast, DDoS is highly homogeneous in its destination port frequency distribution and has significantly higher volume associated to them, mostly concentrated in 1883 (MQTT) and 80 (HTTP).

4.4 Feature Correlation Results

4.4.1 IoT

The correlational analysis for the IoT dataset showed temporal related features as the ones with higher positive correlation. Several feature pairs exhibited high Pearson correlation coefficient ($r > 0.97$), indicating that they might capture the same information, with almost identical variation across the dataset.

The features `Flow IAT Max`, and `Idle Max` present a near-perfect correlation ($r > 0.99$) and similar strong correlations was also observed among Idle statistics (`Idle Min`, `Idle Mean`, and `Idle Max`), as well as among Fwd IAT features, (`Fwd IAT Mean` and `Fwd IAT Min`), reinforcing the consistent pattern behavior involving temporal features.

The correlation heatmap 4.3 illustrates how Idle and IAT statistical features form a strongly correlated group, indicating that these features might capture repetitive time aspects in network activity. Additionally, the correlation between the features `Fwd IAT Total` and `Flow Duration`; and `Fwd IAT Max` and `Flow IAT Max`, both with $r > 0.98$ further supports strong dependency among temporal features.

Although high correlation does not automatically imply redundancy, the result shows clear information overlap and potential redundancy that must be analyzed further on, taking into consideration each feature's singularities.

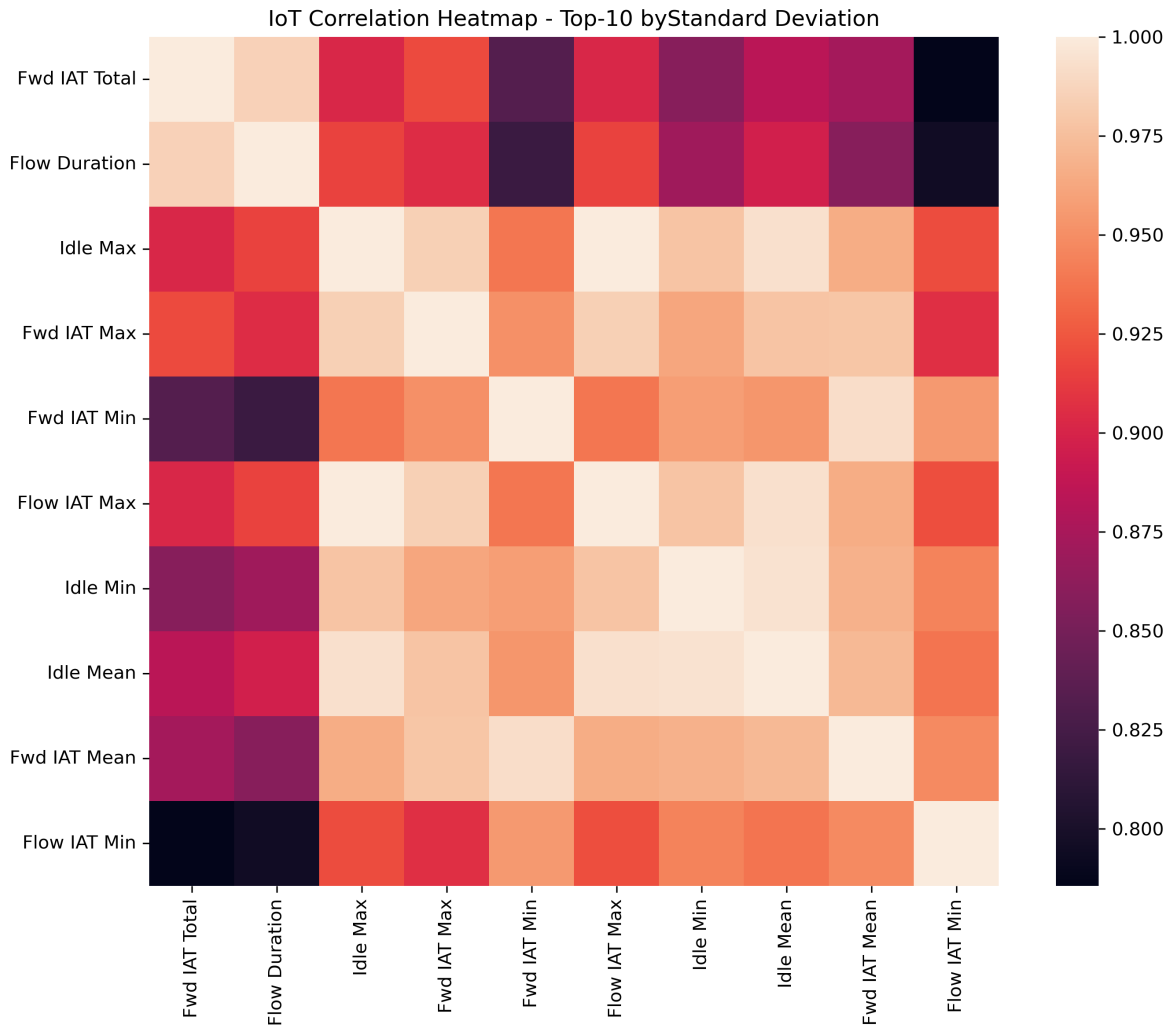


Figure 4.3: IoT Correlation Heatmap - Top-10 by Standard Deviation

4.4.2 Industrial IoT

The analysis of feature correlation for the IIoT dataset showed high association between structural traffic features. Some feature pairs such as `network_packets_dst_count` and `network_packets_all_count`, as well as `network_ports_all_count` and `network_ports_src_count` presented high correlation coefficient ($r > 0.99$). Additionally, the same near-perfect correlation was computed for TCP flag features such as `network_tcp-flags-rst_count` and `network_tcp-flags-fin_count` ($r > 0.99$).

Window size metrics remained highly correlated across statistical representation of features, such as `network_window_size_max` and `network_window_size_std_deviation` reaching $r = 0.98$, as well as `network_window_size_avg` with both `network_window_size_std_deviation` and `network_window_size_max`, reaching $r = 0.97$.

For port count and packet count features the correlation remained moderately high ($r = 0.77$). In general, the heatmap of 4.4 highlights the clustering among packet volume, port usage,

TCP flags, and window size attributes, which represent structural traffic characteristics of the IIoT dataset, suggesting strong intra-domain dependencies.

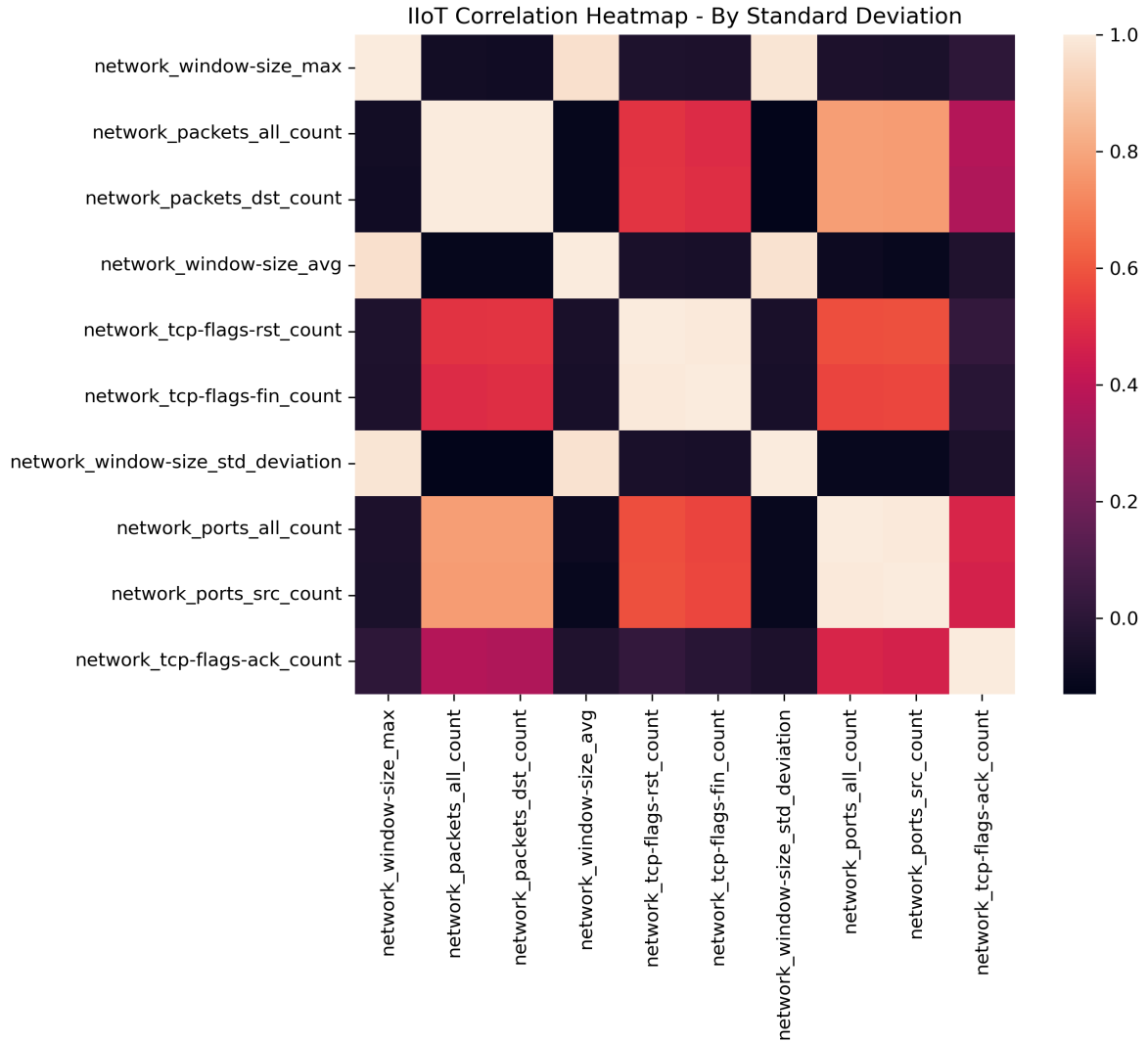


Figure 4.4: IIoT Correlation Heatmap - Top-10 by Standard Deviation

4.5 Feature Behavior Across Structural Categories

4.5.1 IIoT

Protocol dependent features were identified by comparing statistical behavior between features common for both protocols, TCP and UDP. The dataset contains 3,236,935 TCP flows and 625,349 UDP flows, corresponding to 83.80% and 16.20% respectively.

In addition, descriptive statistics shows that temporal features present the largest differences between TCP and UDP. The highest difference in protocol's variability was computed for Flow IAT Min (3.18×10^7), followed by Flow IAT Mean (3.01×10^7). Furthermore, results shows

additional features presenting differences of the same order of magnitude: Fwd IAT Min, Fwd IAT Mean, Idle Min, Idle Mean, Fwd IAT Max, Idle Max, and Flow IAT Max.

The variability difference between protocols is further illustrated in 4.5. Features with higher variability in TCP are represented by positive values and those with higher variability in UDP are represented by negative values.

While the analysis consisted of comparison of common features for both protocols, and the dataset does not indicate features that are exclusive for one protocol, a subset of features presented no variation within one or both protocols.

This behavior was observed for flag-related features, such as PSH, URG, FIN, SYN, RST, ACK, and ECE indicators, as well as for bulk-related metrics, including Fwd Bytes/Bulk Avg, Fwd Packet/Bulk Avg, and Fwd Bulk Rate Avg. Window-based features, such as FWD Init Win Bytes and Bwd Init Win Bytes, and Fwd Seg Size Min also present constant zero values.

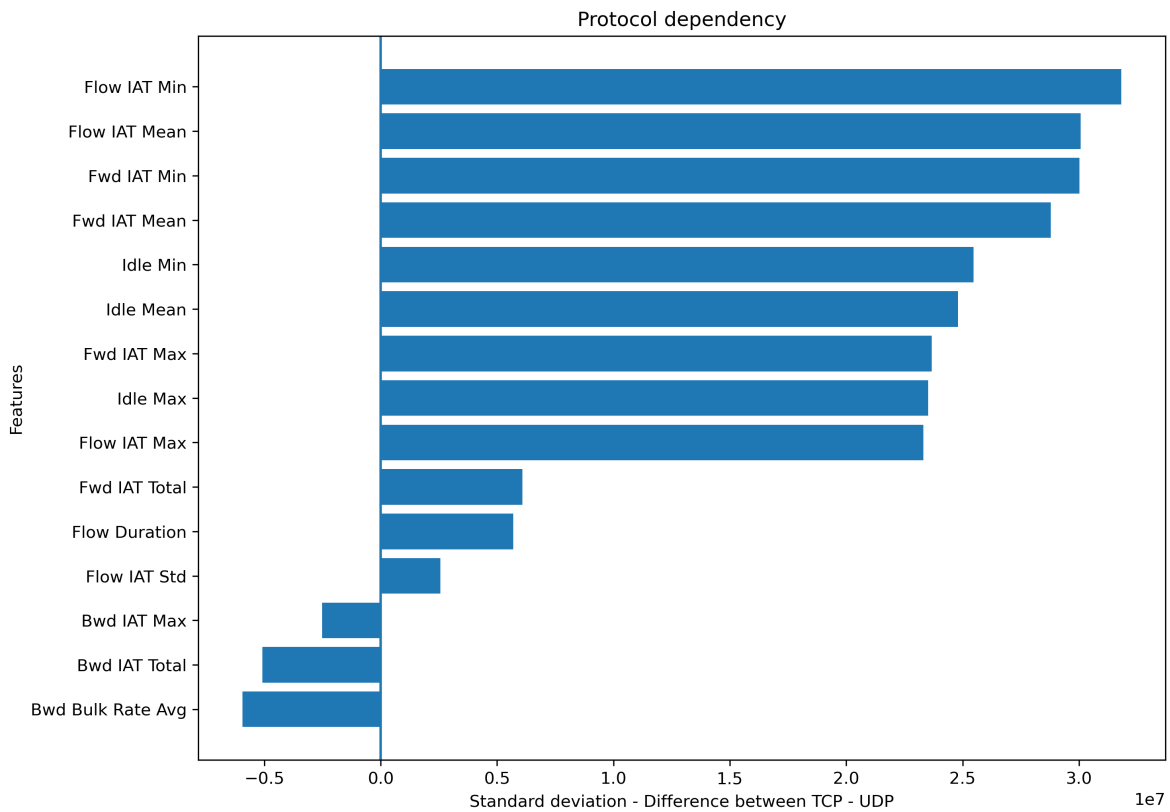


Figure 4.5: Standard deviation - Difference between TCP - UDP

4.5.2 Industrial IoT

The features were grouped according to their indication in the features description, and a group-level analysis revealed Address Diversity group with the highest average discriminative

power in the IIoT dataset, with mean value for $RBC = 0.716$ and low variability computed for features from this group ($std = 0.079$), indicating a consistent high discriminative power within this group. Packet Traffic Rate presented the second highest average RBC ($mean = 0.624$), however, with considerable higher variability than the one observed in Address Diversity group ($std = 0.257$), indicating that features from this group do not evenly discriminate between traffic class. Network Multiplexing also presented relatively high discriminative power ($RBC_{mean} = 0.517$; $std = 0.161$).

The groups Size/Length, Timing Control, and Header Flags exhibited low discriminative power ($RBC < 0.291$) and low variation ($std < 0.178$). As shown in 4.6, these features presented a consistently lower discriminative power than observed in the first half.

Overall, the discriminative power is not uniformly distributed across feature groups, as perceived from the average RBC computed for each group. The following subsection provides a feature-level evaluation of discriminative power.

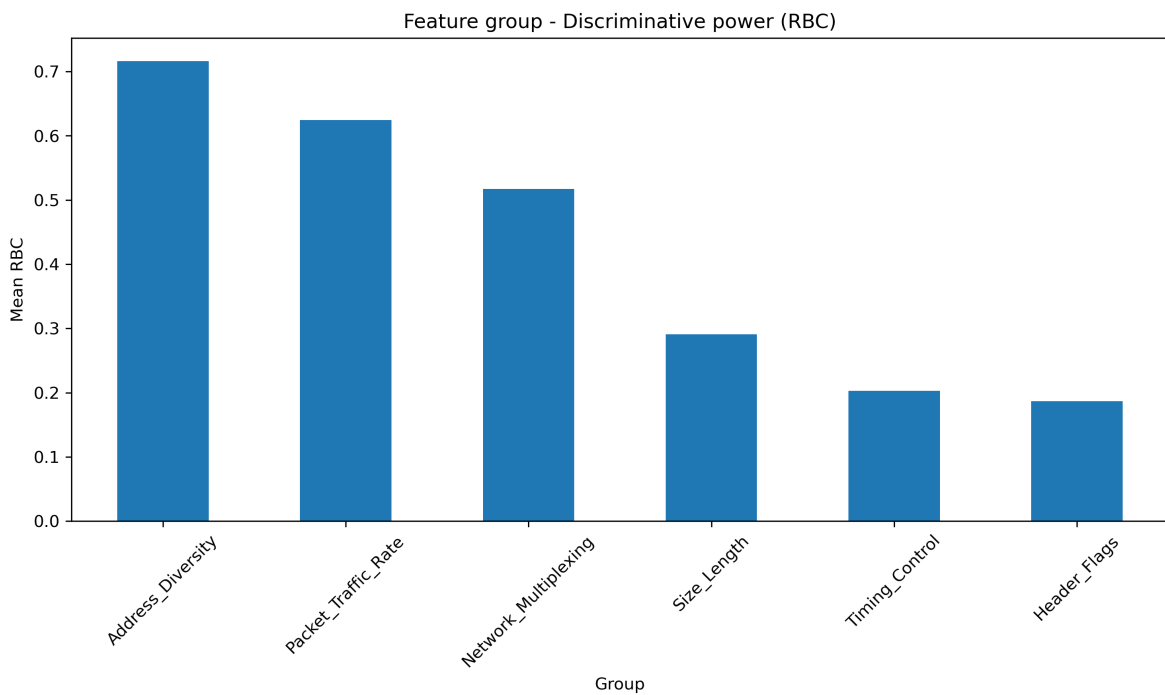


Figure 4.6: Discriminative Power Across Groups

4.6 Discriminative Power Results

4.6.1 IoT

The strongest discriminative power was observed in packet size/length and time-behavior features. Four features presented large effect strength: Packet Length Mean ($RBC = 0.579$), Average Packet Size ($RBC = 0.579$), Packet Length Max ($RBC = 0.534$), and Flow IAT Min ($RBC = 0.503$).

For these features, the median observed in attack traffic was consistently higher than in benign traffic, indicating that in DDoS flows there are more repetitive and uniformly structured communication patterns compared to benign traffic.

Additionally, traffic intensity and flow behavior features presented moderate discriminative power. Features such as Flow Packets/s ($RBC = 0.390$) and Fwd Packets/s ($RBC = 0.389$) presented clear statistical inferences between classes, reflecting the presence of variations in packet transmission rates. Similar discriminative power was also exhibited in Idle min, mean and max features ($RBC \approx 0.340$), highlighting different pause dynamics across traffics.

At least 50% of the numerical features analyzed exhibited a small effect size ($0.33 \leq RBC \leq 0.34$), showing limited discriminative power. Base features such as Flow Duration ($RBC = 0.187$) and Flow Bytes/s ($RBC = 0.133$) consistently showed statistical differences, but in a global context, they have weaker discriminative consistency.

Finally, very small effect size ($RBC < 0.10$) mostly detected TCP flag features. These features presented minimal or absent variation, providing very limited contribution for distinguishing between traffic classes.

Overall, the results indicate a high discriminative power in distinguishing benign from DDoS traffic in features that capture packet size distribution, temporal dynamics, and traffic intensity, while TCP flags, previously indicated as features without variance, presented highly limited ability to differentiate between benign and DDoS traffic.

4.6.2 Industrial IoT

The strongest discriminative power was observed in packet count and network address related features. The features `network_packets_dst_count` ($RBC = 0.845$), `network_packets_all_count` ($RBC = 0.841$), and `network_macs_all_count` ($RBC = 0.751$) presented large effect sizes, with median values equal to 0 for benign traffic and substantially higher values for DDoS traffic. The features `network_ips_src_count` ($RBC = 0.744$) and `network_ips_all_count` ($RBC = 0.743$) presented similar behavior, indicating association of DDoS traffic with higher diversity and intensity of network interactions.

High discriminative power was also computed for port-related features, particularly `network_ports_all_count` ($RBC = 0.652$) and `network_ports_src_count` ($RBC = 0.646$), reinforcing communication multiplicity as a relevant characteristic in attack traffic. Additionally, `network_packet_size_min` ($RBC = 0.645$) showed high discriminative capability, with consistently higher values observed in DDoS traffic.

Next, `network_protocols_dst_count` ($RBC = 0.591$) and `network_ports_dst_count` ($RBC = 0.550$), also presented large effect sizes, as they address protocol diversity and destination

behavior, this result highlights how communication endpoints are structured across traffic classes.

Features such as `network_time-delta_min` ($RBC = 0.489$), `network_ttl_min` ($RBC = 0.478$), and `network_ip-length_min` ($RBC = 0.469$) presented moderate discriminative power, while showing consistent different behavior between benign and attack traffic. However, compared to features with larger effect size, they have greater overlap between benign and attack distributions, indicating a reduced ability to clearly separate the two classes.

As for TCP flag features, the results are highly heterogeneous. While `network_tcpflags-rst_count` ($RBC = 0.445$) and `network_tcp-flags-ack_count` ($RBC = 0.354$) showed moderate discriminative power, the greatest portion of flag-related features presented small to very small effect sizes, indicating limited contribution to class differentiation.

4.7 Feature Selection

The following subsection presents the feature selection methodology adopted in this research, where relevant features for DDoS detection are identified by combining statistical analysis, correlation analysis, and insights from the existing literature.

4.7.1 IoT Features

Flow-Based and Core Traffic Features `Flow Duration` presented a highly skewed distribution, with the mean (3.47×10^7) higher than the median (1.08×10^7). When comparing `Flow Duration` across each traffic class, DDoS traffic showed consistently higher measures of central tendency, indicating longer flows for attack traffic.

However, minimum and maximum values were identical for both traffic classes, reflecting the small effect size ($RBC = 0.187$) and demonstrating that statistical results for this dataset are not favorable for `Flow Duration` to be a reliable feature to distinguish between traffics.

In contrast, literature reinforces that this feature is a strong indicator of malicious activity. As in Sayed et al. (2022) especially DDoS in Software Defined Networking (SDN), due to centralized control network behavior. Despite its relevance in the literature, `Flow Duration` was excluded from the selected feature set due to its low discriminative power and limited contribution to class differentiation in this dataset.

Correlation was also computed for `Flow Duration`, `Flow Bytes/s` and `Flow Packets/s`, which represent time, intensity and frequency, respectively. Results show almost complete independence between these features. When comparing intensity and frequency, `Flow Bytes/s` ($RBC = 0.133$) and `Flow Packets/s` ($RBC = 0.390$), packet transmission frequency showed higher discriminative power.

Flow Bytes/s, indicates low discriminative power to differentiate between traffic classes. While benign traffic presented a higher median compared to DDoS traffic, the feature exhibited high dispersion and extremely high maximum values. These characteristics indicate a wide variability and overlapping distributions between traffic classes. Therefore, mathematical foundation supports this feature to not be included in the reduced dataset.

The feature Flow Packets/s was retained due to its ability from differentiating between traffics and the representation relevance: the number of flow packets per second, capturing traffic intensity over time. Central tendency measures were also compared, showing that the median and upper quartile (14.23 and 31.72) are consistently higher in benign traffic than in DDoS traffic (0.32 and 16.50).

Directional and Packet Length Features In order to capture directional behavior in network communication, we first analyzed Total Fwd Packet and Total Bwd packets. Although the Total Bwd packet presented a moderate effect size ($RBC = 0.425$), the Total Fwd packet showed a smaller effect size ($RBC = 0.175$). When combined with their low correlation ($r = 0.14$), these results reflect the nature of network communications, where benign traffic involves bidirectional exchanges, while DDoS traffic is characterized by high intensity in the forward direction with limited or absent response from the destination. Therefore, Total Bwd Packet was retained for further analyses.

Packet length features were further analyzed considering both forward and backward directions. Results showed that forward and backward features capture distinct aspects of traffic behavior, as indicated by their low correlation. In general, packet length in the backward direction presented higher discriminative power than forward features. This is explained by the nature of the traffic: forward packets represent attacker-generated traffic, while backward packets reflect the system response. In this sense, system responses tend to present more stable patterns, allowing backward features to better discriminate between traffic classes.

An exception was observed for Fwd Packet Length Min ($RBC = 0.327$), which is consistent with both statistical and literature insights findings. This suggests that small and consistent packet sizes in the forward direction are strong characteristics of DDoS attacks, where attackers aim to maximize packet transmission while minimizing resource consumption, generating a large number of lightweight packets (Zhou et al., 2017).

Backward packet length features were also analyzed separately. High correlation ($r > 0.84$) was observed between Bwd Packet Length Min, Bwd Packet Length Mean and Bwd Packet Length Max. In contrast, Bwd Packet Length Std presented weak to moderate correlation with the other features in the same group (ranging from $r = -0.058$ to $r = 0.452$), indicating that it captures a different aspect of the data. While Min, Mean and Max describe the magnitude of backward packet length, Bwd Packet Length Std reflects how the diversity in packet length within the flow. Among the packet length magnitude features, Bwd Packet Length

Max was selected, as it presented the highest discriminative power ($RBC = 0.352$). Additionally, despite its limited discriminative power ($RBC = 0.205$) Bwd Packet Length Std was retained, as it provides additional information that might not be captured by central tendency measures.

Finally, Total Length of Bwd Packet consistently presented low correlation with other features from this subgroup and highest ($RBC = 0.358$) and therefore, it was retained.

This distribution aligns with findings in the literature, as in Gu et al. (2020), where high-rate DDoS attacks differ from low-rate attacks, commonly observed in IoT environments. While high-rate attacks are characterized by large packet volumes, low-rate attacks aim to evade detection by reducing the packet rate, making the difference from legitimate traffic less noticeable.

Traffic Intensity and Directional Rate Features Previously, the baseline indicator of traffic frequency, Flow Packets/s, was analyzed and retained. To further investigate packet transmission rates across flow, the investigation now examines the traffic frequency according to the directionality of the flow of the packets.

Both Fwd Packets/s and Bwd Packets/s features presented moderate discriminative power. In a closer inspection, benign traffic presented consistent high rates in both directions, reflecting the bidirectionality of the communication. In contrast, DDoS traffic presented lower values and strong concentration of zero values in the backward packet rates, another confirmation that in DDoS traffic have limited or absent systems response. Therefore, Fwd Packets/s and Bwd Packets/s were retained because they effectively capture relevant and different aspects of communication patterns.

Temporal Features – Inter-Arrival Time (IAT) IAT features describe the time intervals between consecutive packets and provide relevant information about bursty and uniform packet transmission.

Strong correlation ($r > 0.92$) was observed for Flow IAT Min, Flow IAT Mean and Flow IAT Max, indicating redundancy within this group. Among them, Flow IAT Min presented the highest discriminative power ($RBC = 0.503$). This suggests that the lowest interval between two packets in the flow is a stronger characteristic of DDoS traffic. This behavior is aligned with literature, where network traffic exhibits irregular and bursty temporal patterns rather than uniform packet transmission (Garsva et al., 2014).

Additionally, Flow IAT Std showed no correlation with the other features of this group, but brings a global overview of how these features are distributed. While Min, Mean and Max describe magnitude-related aspects of inter-arrival time, the standard deviation reflects how these values vary within the flow. Despite the reduced effect size ($RBC = 0.305$), it is still

considered moderate. Therefore, Flow IAT Min was retained as the primary feature and Flow IAT Std as a complementary feature.

Among Fwd IAT features, high redundancy was also observed ($r > 0.95$). Fwd IAT Std presented the highest discriminative power ($RBC = 0.310$) and was therefore retained. Although statistics computed for each traffic class in Fwd IAT Std showed a high concentration of zero values in the lower quartile and median, the upper quartile remained zero for DDoS and significantly higher for benign traffic. This explains the highest RBC observed for this feature, as it captures differences in temporal dynamics, rather than only measuring sparsity in inter-arrival time.

In contrast to forward features that reflect attack behavior, backward IAT features indicate the system response. The features of the backward IAT group consistently presented low discriminative power to differentiate between traffic classes. Therefore, all features of this group were excluded.

Activity and Idle Features Temporal features related to activity and idle periods were analyzed to capture flow-level behavior in terms of pauses and activity of the flow. Despite both Active and Idle being time representation features, they computed weak negative correlation ($r \approx -0.05$), highlighting independence between both groups.

All Active features presented similar results within each traffic class, with the presence of extreme outliers elevating the mean values for benign traffic. This suggests that benign flows tend to remain active for longer periods, while DDoS traffic shows more constrained active behavior. Due to the similarity among Active features and their limited differentiation between traffic classes, these features were excluded.

The features Idle Min, Idle Mean, and Idle Max presented near-perfect correlation ($r > 0.99$), indicating strong redundancy within this group. Additionally, previous analysis of time-related features across the dataset also pointed out high correlation among Idle features.

While IAT captures information about time between packets, Idle represents longer inactivity periods before becoming active. In other words, every Idle involves IAT, but not every IAT can be considered an Idle. In benign traffic, short pauses tend to become idle periods, contrasting with DDoS traffic, where temporal patterns are more uniform.

Due to the high correlation between features of this group and their relationship with other temporal features, only Idle Min was retained, as it provides a representative measure of inactivity periods within the flow.

Header and Segment-Based Features Clear difference between traffic class were found in Header length features, with DDoS traffic consistently exhibiting lower mean values, while benign traffic presented greater variability. This suggests that benign traffic involves more

heterogeneous communication patterns, while DDoS traffic is more limited in terms of header usage.

Segment size features, including Fwd Segment Size Avg, Bwd Segment Size Avg and Seg Size Min, were analyzed within the same group. The correlation between these features ranged from $r = -0.05$ to $r = 0.436$, indicating weak to moderate relationships and suggesting that they capture partially distinct aspects of the transmission structure.

Fwd Segment Size Avg showed clear differences in median between traffic classes. However, it also presented similar maximum and shared minimum values for both traffics, limiting the discriminative power. In contrast, Seg Size Min consistently presented different statistics for each traffic class ($RBC = 0.349$), indicating stronger discriminative capability.

Bwd Segment Size Avg reflects system response to the traffic, and its distribution shows that under attack there is consistently smaller segment size in the backward direction. This behavior is strongly related to the bidirectional nature of network communication and how the system responds when under attack.

In general terms, results suggest that DDoS traffic tends to present smaller segment sizes compared to benign traffic, indicating differences in traffic structure. However, despite individual observations suggesting that these features are relevant indicators for distinguishing between traffic classes, their discriminative power remains limited according to RBC results

Therefore, these features were not retained, as their contribution is reduced when analyzed in conjunction with other features.

Flag-Based Features (TCP Flags) TCP Flag Count features were analyzed to capture connection control behavior in network communication. Directional features, including Fwd PSH flags, Bwd PSH Flags, Fwd URG Flags, and Bwd URG Flags, represent the number of times these flags were observed in packets travelling in the forward and backward directions. However, these features presented extremely low variation across dataset, with most observations around zero. Therefore, they do not contribute to class differentiation.

TCP Flag Count features presented limited discriminative power overall, according to their low RBC values. URG Flag Count, CWR Flag Count and ECE Flag Count were almost always absent, while PSH Flag Count and FIN Flag Count concentrated most of observations at zero. These features have extremely limited or nonexistent discrimination power.

RST Flag Count is the only feature with negative correlation with other features from TCP flag group. The results indicate that it captures a distinct aspect of connection control behavior.

Unlike the remaining TCP flag features, the results indicate that RST Flag Count captures a different aspect of connection control behavior associated to DDoS attack.

Other studies based on the same dataset produced similar results when calculating feature importance, such as in Mgungile and Tonkal (2025) where RST Flag Count was selected using an automated ANOVA test.

Therefore, RST Flag Count was retained, while the remaining features of the TCP flag were excluded due to low discriminative power and redundancy.

Bulk-Based Features Bulk related features capture extremely rare events, with all statistical measures equal to zero.

Subflow and Window-Based Features Subflow and window-based features were analyzed to capture flow division and window control behavior in network communication. Subflow Fwd Bytes ($RBC = 0.447$) Subflow Fwd Packets ($RBC = 0.321$) presented moderate discriminative power, therefore were retained. The same features in backwards direction presents small to very small RBC effect size.

FWD Init Win Bytes represents the total number of bytes sent in the initial window in the forward direction. Although this feature presented limited discriminative power ($RBC = 0.264$), it still highlights a characteristic of DDoS attacks, where traffic often presents repetitive and automated patterns, including its network structure (Munshi, 2024). However, despite the relevant structural characteristics of DDoS traffic, this feature presented a limited ability to clearly distinguish between traffic classes.

When compared to Bwd Init Win Bytes ($RBC = 0.052$), FWD Init Win Bytes provides more information about how the attack occurs rather than how the system reacts when under attack.

Therefore, Subflow Fwd Bytes, Subflow Fwd Packets and FWD Init Win Bytes were retained, while the remaining features in this group were excluded due to their limited contribution.

Payload Feature Fwd Act Data Pkts is related to payload-based traffic characteristics and measures the number of data carrying actual data client-server direction. However, results showed that this feature cannot distinguish well between traffics ($RBC = 0.059$). In this sense, according to Finsterbusch et al. (2014), payload-based classification is fundamentally limited, especially in modern network environments where encryption and application-layer attacks, such as HTTP Flood are common. As result, pay-load-based features are substantially reduced in their discriminative power.

4.7.2 Industrial IoT Features

Size/Length Features Header lenght features presented high discriminative power ($RBC \approx 0.052$), except for `network_header-length_std_deviation`, due to the lack of variability intra-

flow, resulting in standard deviation close to zero in both classes.

The correlation among Size/Length features showed that they are nearly perfectly correlated, suggesting redundancy between them. In addition, the average representation was selected because it is more stable than the minimum and maximum, where extreme values may reduce interpretability. Therefore, `network_header-length_avg` was retained, while the remaining features were excluded due to redundancy and lower interpretability.

The same limited discriminative power was computed for `network_iplength_std_deviation`. However, results of correlation calculated for IP length features showed that they capture distinct aspects of network structure. Therefore, `network_ip-length_max` and `network_ip-length_avg` remained for further analysis since they inform, respectively, extreme values of packet size and common IP packet length in a time window.

Despite the moderate RBC effect strength of `network_ip-length_min`, this feature has lower interpretability compared to the others features of the same group. As for `network_ip-length_std_deviation`, this feature does not have statistical significance for distinguishing traffics. Thus, only `network_ip-length_max` and `network_ip-length_avg` were retained.

In TCP communication, Maximum Segment Size (MSS) specifies how large data can be sent in one segment without fragmentation, excluding the lengths of the TCP and IP headers (Alcock & Nelson, 2010). Therefore, MSS features reflect more about network configuration than traffic behavior, since these parameters are pre-established.

Statistical analysis also revealed near-perfect correlation between these features, indicating the lack of meaningful variability. This is also confirmed by RBC effect strength results, which show small discriminative power for these features.

Packet size features represent the actual size of packets transmitted in time window, therefore, reflect communication patterns that are relevant for distinguishing between benign and attack traffic.

Among their statistical representations, `network_packet_size_min` has the highest discriminative power ($RBC = 0.646$). Additionally, a concentration of low values was observed in benign traffic, with median equal to zero, and upper quartile consistently smaller compared to the lower quartile, mean and upper quartile in DDoS traffic.

The correlation was also computed for packet size features, in which it was demonstrated that they are not highly correlated and thus not redundant.

The feature `network_packet_size_avg` was also selected for the formation of the reduced dataset because it has a moderate RBC effect strength and the lack of strong correlation of similar features provides complementary information to each other. The features `network_packet_size_max` were discarded despite its moderate RBC effect size, because when they were

individually analyzed, the distribution revealed to be highly dependent on extreme values, indicating outlier influence on the results. As for the `network_packet-size_std_deviation`, median is close to zero for both classes, which results in a small RBC effect size, therefore, this feature was excluded.

The strongest correlation computed for the payload length features was between the `network_payload-length_max` and the `network_payload-length_std_deviation` ($r = 0.92$). In general, payload length features indicate that they capture distinguishing behavior patterns.

Additionally, `network_packet-size_avg` and `network_payload-length_avg` are highly correlated ($r = 0.96$).

Interestingly, packet size features ranked at the top in RBC effect strength, while all payload Length features presented small RBC effect size. This is justified by the fact that for both traffic classes the median values are similar, leading to small effect size, despite their strong correlation with packet size features.

Therefore, the entire set of features will be discarded for further analyses due to their proved statistical weakness and the fact that there are other features that already capture similar aspects, but with strong discriminative power.

Thus, all Payload Length features were excluded.

Packet Traffic Rate Correlation computed for this group revealed a near-perfect correlation between `network_packets_all_count` and `network_packets_dst_count` ($r > 0.99$), suggesting overlapping information.

`network_packets_all_count` represents the total number of packets observed in a time window, therefore reflecting traffic volume characteristics.

The RBC effect size for this feature is the second largest ($RBC = 0.841$) and statistical analysis also revealed this feature rank second in terms of standard deviation. The benign traffic has very low packet counts, often close to zero, while the attack traffic presents a more consistent pattern with high packet volume. The results match the literature, where many studies confirm the relation between DDoS attacks and large traffic volumes generated in short periods of time. Therefore, `network_packets_all_count` was not only retained for the reduced dataset, but also considered one of the most important features in the IIoT dataset.

The feature `network_packets_src_count` showed moderate RBC effect size and meaningful statistical results for distinguishing between traffic. While benign traffic has most values concentrated at zero, attack traffic presents higher median and upper quartile values. Additionally, it has low correlation with `network_packets_all_count` ($r = 0.26$), indicating that they capture different information. In contrast to the total number of packets,

`network_packets_src_count` provides complementary information about traffic behavior and was retained for further analysis.

The feature `network_interval-packets` represents the time interval of packets in a time window. Even though the dataset collects information in a 1-second window, literature shows that network traffic is bursty, especially in shorter periods than a full second. According to its RBC effect size ($RBC = 0.351$), this feature presents moderate discriminative power to differentiate between benign and DDoS traffic.

Header Flags All features from the Header Flags group presented small to very small discriminative strength to differentiate benign and attack traffic. The values for IP flags are limited due to their binary structure where: bit 0 is reserved and must be zero, bit 1 indicates Don't Fragment (DF), and bit 2 indicates More Fragments (MF) ("Internet Protocol", 1981). These flags only describe how packet fragmentation is handled in the traffic and, combined with their low variability and low discriminative power, do not provide relevant information for class separation in this dataset. Therefore, all IP flag features were discarded in the next step of the analysis.

For TCP flags, the correlation analysis showed substantial redundancy among features. ACK and PSH counts were highly correlated ($r = 0.98$) while FIN and RST counts presented a near-perfect correlation ($r = 0.98$). In addition, the statistical representations of the TCP flags were also strongly correlated ($r = 0.87$), and URG did not show variability, therefore it was not included in the reduced dataset.

Interestingly, from a theoretical perspective, PSH is associated with immediate data delivery to the application layer, which is relevant in real-time environments such as IIoT systems (Jerry Sheehan, 2025). However, in this dataset, its statistical behavior did not support a meaningful discriminative value. Although highly correlated with ACK, PSH remained practically identical across both classes, with median and upper quartile values equal to 0 and 1, respectively. This contrasts with ACK flags, whose median and upper quartile were considerably higher in attack traffic. Therefore, despite its theoretical relevance, PSH did not provide consistent discriminative power in this dataset and was excluded.

Similar comparison results were observed between the FIN and RST flags. Both are related to connection termination, which is supported by their near-perfect correlation. However, RST is more strongly associated with abrupt connection termination, which is more consistent with attack behavior. This interpretation is supported by the moderate RBC effect size for RST, compared with the very small effect size for FIN. Thus, RST was retained, while FIN was excluded.

Regarding SYN flags, they are classically associated with SYN flood attacks, a well-known DDoS pattern. However, in this dataset, the use of a 1-second time window likely reduces

its discriminative capacity, since SYN flood behavior is better captured through the accumulation of incomplete connections over time, especially through the imbalance between SYN requests and ACK responses (Bogdanoski et al., 2013; Vinicius Fulber-Garcia & Michal Aibin, 2024). As a result, SYN presented only a small RBC effect size. Even so, it was retained due to its recognized relevance in the literature and because its limited performance can be reasonably explained by the time-window structure of this dataset.

Finally, the statistical representations of TCP flags — such as mean, minimum, maximum, and standard deviation — were excluded. Although some isolated descriptive results suggested differences between traffics, these patterns were not supported by the overall RBC analysis. Since these aggregated measures were also highly correlated and specific TCP flag counts had already been retained, they were considered redundant and not kept in the reduced dataset.

Importantly, these findings should be interpreted as specific to this dataset and to its feature extraction strategy. The low discriminative power observed for several header flag features does not necessarily mean that these variables are irrelevant in other studies, since their usefulness may vary depending on traffic characteristics, attack types, and especially the time window adopted for feature computation.

Address Diversity The feature `network_ips_all_count` represents the total number of IP addresses observed in a time window. This feature is statistically strong for distinguishing between benign and attack traffic, presenting median consistently higher in the attack sample (6), while this value is 0 for benign traffic. Additionally, the large RBC effect strength confirms that this is a meaningful feature for further analysis. Literature also highlights that IP addresses are widely used to detect malicious activities, since attacks often involve multiple IPs (Niakanlahiji et al., 2017).

Network IP features are also highly correlated. Correlation computed for this group revealed very high correlation coefficients across all IP-based features, indicating redundancy. However, `network_ips_all_count` was retained as representative of network IP features due to its strong repdiscriminative power.

The feature `network_macs_all_count` has near-perfect correlation with `network_ips_all_count`. Despite both features capturing similar characteristics, IP features reflects communication patterns, such as network interactions and data exchange. On the other hand, MAC addresses are related to hardware and are more useful for device identification (Garg et al., 2012). Therefore, only `network_ips_all_count` was retained, while all network MAC features were excluded.

Timing Control The correlation matrix from this group revealed different behaviors associated with each feature subgroup. Time delta and TTL features presented lower correlation levels compared to window size features.

Further on, the RBC effect strength was analyzed. Among features of the time delta subgroup, `network_time-delta_min` presented higher discriminative power ($RBC = 0.498$), while `network_time-delta_std_deviation` showed a smaller ($RBC = 0.204$) but complementary effect, since there is a negative correlation between these features ($r < 0$). Therefore, both were retained. As for `network_time-delta_max` and `network_time-delta_avg` showed a very small effect of RBC, therefore they were excluded.

Additionally, both `network_ttl_min` ($RBC = 0.479$) and `network_ttl_max` ($RBC = 0.339$) showed moderate discriminative power. Other features from the same subgroup presented very small RBC effect size.

All TCP window size features showed low discriminative power ($RBC \leq 0.234$) and high redundancy, justifying their exclusion.

These findings are consistent with the protocol nature of the features. TTL-based features are derived from IP headers and reflect network-layer properties, influencing structural differences in traffic behavior. In contrast, TCP window size features are associated with flow control mechanisms of the transport layer, which are highly dependent on network configuration, limiting their relevance in distinguishing traffic.

Network Multiplexing The Network Multiplexing group captures how communication is distributed across ports and protocols within a time window.

A near-perfect correlation was observed between `network_ports_all_count` and `network_ports_src_count` indicating strong redundancy. `network_ports_all_count` was retained both because of its general ability to represent port features and because it is slightly higher in RBC effect size.

In contrast, `network_ports_dst_count` presented low correlation with other port features. Given the nature of the dataset, multiple destination ports can be observed in each event, meaning that this feature reflects how diverse destination ports can be, instead of precisely informing the traffic volume associated with each individual port. Since `network_ports_dst_count` is complementary to `network_ports_all_count` in the sense that it addresses port diversity, the feature was also retained.

Protocol-related features capture the diversity of communication protocols, as the dataset contains a wide variety of protocols, including TCP, UDP, MQTT, HTTP, TELNET and others. However, most events capture only 0 to 2 protocols. Even with this variability limitation, attack traffic shows a consistent increase in the number of destination protocols, resulting in a large strength of the RBC effect.

Additionally, high correlation was observed for protocol-related features ($0.75 < r < 0.99$), indicating that they capture related aspects of protocol behavior within a time window, but

are not identical. Therefore, `network_protocols_all_count` was removed due to overlapping (not completely redundant) information, and `network_protocols_src_count` due to its lower discriminative contribution. For this set of features, `network_protocols_dst_count` was retained.

The datasets with reduced features captures distinct aspects of network behavior, preserving meaningful information, while minimizing redundancy and less informative features. As a result, the final reduced dataset provides a more efficient and informative representation of traffic patterns, supporting further evaluation in machine learning models.

4.8 Model Performance for Feature Selection Validation

This section presents the performance of the baseline machine learning models described in Chapter 3, in which both IoT and IIoT reduced datasets were used.

The table 4.5 presents the model evaluation results. In general, all scenarios resulted in high performance, indicating that the previously selected features indeed capture relevant network information for the identification of DDoS attacks.

For the IoT dataset, the highest accuracy (0.94) was achieved by the Random Forest model, with consistently high values for precision (0.95), recall (0.98), and F1-score (0.97). While the SVM model showed lower accuracy (0.90), it also presented a higher recall (0.99), resulting in an F1-score of 0.95. Overall, SVM achieved higher recall in attack detection, while Random Forest provides a more balanced performance.

These results can be associated with the nature of the dataset itself, which is highly imbalanced (Zughrat et al., 2014). In this sense, Rui et al. (2025) compared both models used in the present work and presented similar perspective, indicating that although SVM can achieve higher detection rates, Random Forest tends to provide more stable and reliable performance, particularly in reducing false positives.

Consistently with the results obtained from the IoT dataset, the IIoT dataset also showed that the selected features highlight important characteristics for network behavior classification. High accuracy was observed for both SVM (0.97) and Random Forest (0.98). Additionally, Random Forest presented slightly higher evaluation metrics accross all parameters.

Table 4.5: Performance values for IoT and IIoT models.

IoT				
Model	Accuracy	Precision	Recall	F1-Score
SVM	0.90	0.91	0.99	0.95
Random Forest	0.94	0.95	0.98	0.97
IIoT				
Model	Accuracy	Precision	Recall	F1-Score
SVM	0.97	0.92	0.86	0.89
Random Forest	0.98	0.99	0.88	0.93

4.9 Discussion

The combination of statistical analysis, feature selection and experimental validation through machine learning models allow us to make a cross-domain interpretation of feature behavior, highlighting shared and domain-specific patterns between IoT and IIoT environments.

Although IoT and IIoT datasets have all their features extracted in different ways (flow-based and window-based), some feature categories still demonstrated strong discriminative power common to both domains.

In this sense, traffic intensity and structural features are strong markers of DDoS activity, independently of how and where this information was captured. For IoT, statistical data, confirmed by insights from existing literature, indicate that flow packets and directional packet rates capture essential data about variations in transmission frequency. Moreover, this is one of the only features where both forward and backward directions are important. In this case, we can observe not only an intense traffic flow being generated by the attack, but also a concentration of zeros in the opposite direction, which indicates that in DDoS traffic there are limited or absent system responses. In the IIoT data set, `network_packets_all_count` and `network_packets_src_count` are also indications of traffic intensity, both with relevant discriminative power. Although the total count of network packets is highly correlated to forward traffic, the same pattern of limited system response makes the backward traffic intensity indicate a DDoS attack.

With this, it is possible to infer that high volume and high frequency are common traffic patterns across both domains, regardless of how this information was extracted.

Similarity patterns were also identified in features related to packet size and length. For IoT, features related to packet length ranked at the top for higher RBC effect, presenting higher values and wider dispersion. In IIoT, `network_packet_size_min` and `network_packet_size_avg` proved to be features with relevant discriminative power; however, in this environment they present distinct behavior, marked by greater consistency of high values and lower variance observed in attacks.

However, IoT and IIoT also presented unique characteristics of their respective domains.

In IoT, temporal features (IAT and Idle) are important to characterize traffic behavior. These features present high variability, strong skewness and large differences between mean and median, indicating irregularities and burst patterns over time. However, not all temporal features in the dataset have the same relevance. As shown in 4.4, in IoT they dominate the features with higher correlation, although not all have the same power to discriminate between classes.

In contrast, timing features in IIoT presented a more heterogeneous behavior in terms of discriminative power. Although they show high variability and skewed distribution, many present similar behavior across classes, leading to reduced discriminative power. However, this limitation was not observed in all timing features. For example, TTL-based features are derived from IP headers, reflecting network-layer properties and allowing structural differences between classes to be captured. In contrast, TCP window size features are associated with flow control mechanisms of the transport layer, which are highly dependent on network configuration, limiting their relevance for distinguishing between benign and DDoS traffic.

Overall, although timing features are useful to understand general network behavior patterns, their discriminative power is reduced compared to IoT.

Another important cross-domain difference observed is the distribution of communication endpoints. In IoT, destination ports have limited service concentration. Therefore, although destination ports are the same for both traffic types, the volume associated with DDoS traffic is extremely higher compared to benign traffic. Moreover, the frequency of destination ports is substantially higher in DDoS, indicating that not only is there a preference for target endpoints, but also higher intensity of packets being transmitted.

In IIoT, there was a limitation related to the dataset, which by its own structure has aggregated values, so in a single observation multiple destination ports are displayed. Even so, it was possible to obtain an associated, although not precise, value of traffic volume for each destination port. In this sense, 1883 (MQTT) and 80 (HTTP) showed high traffic intensity, but among the 10 ports identified with aggregated traffic intensity, all of them also presented high values. In this sense, we highlight that unlike IoT, in IIoT the destination ports are mostly different for benign and DDoS traffic. Therefore, considering that the calculation for IIoT does not have the same precision of traffic intensity as IoT, the calculations performed so far indicate a lower dependence on endpoint selection and a stronger association with traffic intensity and structural patterns.

Another difference analyzed, which was limited by the nature of the dataset, was the identification of protocol-dependent features. In IoT there was explicitly this differentiation, making it possible to identify that temporal features vary significantly between protocols, showing greater dependence on the TCP protocol. For IIoT, the dataset does not have explicit protocol identification; however, a group-level analysis was conducted to verify which groups

are more informative for DDoS attack detection. The results showed that address diversity has greater discriminative power, and also lower variability among these features, indicating consistent and reliable behavior in differentiating between classes. In this sense, the Packet Traffic Rate and Network Multiplexing groups also have higher discriminative power; however, greater variability in this group suggests that not all features are actually relevant for traffic differentiation.

Finally, some features presented little or no variation between different traffic classes in both domains. Consequently, these features showed very small effect size and did not contribute to DDoS detection.

The results obtained from machine learning models further supports these findings, where satisfactory performance was obtained in both models, indicating that the selected features effectively identify relevant aspects of DDoS behavior. The positive results reinforces that the features identifies in this project as discriminative are not only theoretically relevant, but also practically effective in DDoS attack identification.

4.10 Guidelines for Feature Selection

Based on cross-domain findings, the following guidelines are proposed:

4.10.1 Guideline 1: Traffic Intensity as a Primary Indicator

For both environments, features that capture traffic intensity characteristics should be prioritized, as they consistently present satisfactory discriminative power in all scenarios.

This guideline is supported by results statistical analysis and model evaluation, where features related to traffic intensity such as packet counts and directional packet rates, presented consistently discriminative power observed in both datasets. In particular, the presence of high traffic in the forward direction combined with the limited or absence of backward traffic (system response) was identified as a strong indicator of the behavior of DDoS traffic.

4.10.2 Guideline 2: Packet Size and Length as Structural Indicators

As attacks tend to be more uniform, packet size and length features are also good indicators of DDoS attack, since they capture structural differences in traffic patterns.

Statistical findings support this guideline, where in IoT dataset packet length and packet size features showed largest effect sizes, while in IIoT the DDoS traffic consistently presented higher values.

4.10.3 Guideline 3: Domain-Specific Behavior of Temporal Features

While IoT traffic is human-driven, therefore irregular and bursty, IIoT traffic is typically more structured and with consistent patterns due to automated machine-to-machine communication. Therefore, in IoT temporal features better capture DDoS traffic.

This guideline is supported by the cross-domain analysis, where temporal features are highly heterogeneous, skewed, and discriminative, while in IoT the two traffic classes presented similar behavior, reducing their effectiveness for classification.

4.10.4 Guideline 4: Destination Port Handling Across Domains

IoT environments are limited to a group of destination ports where there is higher packet transmission intensity in DDoS. Although the comparison with IIoT is not in the same calculation terms, destination ports in IIoT are more diverse, even though there is higher intensity associated with 1883 (MQTT) and 80 (HTTP). Therefore, we recommend that these features be treated more independently in further analyses, for example by applying one-hot encoding.

This guideline relies on the observed difference in endpoints distribution. While IoT concentrates traffic intensity in specific ports, IIoT presented more distributed communication patterns across multiple ports. The different pattern observed in IoT and IIoT indicates that destination port requires domain-specific treatment.

4.10.5 Guideline 5: Importance of Dataset-Specific Validation

Statistical results do not always align with the literature. Therefore, we suggest a manual analysis, even if superficial, of the features. As was the case, for example, with flow duration. Although the literature is categorical in emphasizing the importance of this feature, in the IoT dataset used in this research it was decided that it was not a reliable feature to distinguish between traffic types.

This guideline is supported by the observed inconsistencies between statistical results and expected feature relevance from the literature. In some cases, features considered highly relevant in the literature proved statistically irrelevant in the context of the dataset, reinforcing that some features need dataset-specific validation before feature selection.

In general, the proposed guidelines summarize the main findings and identify similar and unique patterns in the IoT and IIoT environment. The results contribute to a better understanding of feature relevance in different contexts, supporting the development of more adaptable and generalized machine learning-based intrusion detection systems.

The proposed guidelines summarize the main findings and identify similar and unique patterns in the IoT and IIoT environment, improving the understanding of feature relevance

across domain. In general terms, it supports the development of more adaptable and generalized machine learning-based intrusion detection systems.

5

Conclusion

5.1 Introduction

In Chapter 2 we present a set of theoretical foundations about IoT and IIoT networks, highlighting their respective architectures and the most common attacks in these types of networks. Next, in Chapter 3 we present the methodology used to achieve the results of Chapter 4, where the evidence-based guidelines for feature selection in IoT and IIoT networks were also presented. In this chapter, the main results and findings will be discussed regarding the project objectives, research contributions will be presented, as well as future work recommendations, based on the limitations that we faced.

5.2 Summary of Research

This research applied a quantitative and data-driven approach to understand feature behavior across IoT and IIoT environments. However, not only the domain was different, but also the feature extraction and representation, leading the analysis to focus on identifying common and specific discriminative patterns across domains.

The research explored the statistical behavior of CIC IoT DIAD 2024 and CIC IIoT 2025 datasets in order to identify which features provide more relevant information about traffic patterns, especially DDoS traffic. Insights from the literature review combined with statistical

results allowed for a more comprehensive interpretation of the importance of these characteristics. Next, features identified in this process as more relevant were used to create a reduced dataset for each domain.

To validate the theoretical findings, two baseline machine learning models, Random Forest and Support Vector Machine (SVM), were applied to the reduced datasets. The results were satisfactory for both models, with Random Forest performing slightly better for both datasets.

Additionally, despite their differences regarding domain, feature extraction and representation, some feature groups such as traffic intensity and structural network behavior remained consistently relevant for DDoS discrimination.

Overall, the findings supported the creation of evidence-based guidelines for feature selection, while reinforcing that some traffic characteristics remain informative even when network structure and feature representation are different.

5.3 Research Objectives

The objectives defined in Section 1.4 are evaluated as follows:

Secondary Objective 1 – To perform a comparative feature behavior analysis: This objective was addressed through the analysis of both IoT and IIoT datasets. The Findings show that traffic intensity and packet-related features have a similar behavior across domains, as well as temporal features and destination port that behave differently in each domain. Since the comparison parameters are not uniquely associated to a specific section, Chapter 2 provided with related literature, which was combined with statistical inferences from sections, discussed in a higher level of comparison in section discussion, and summarized, in evidence-based guidelines in Section 4.10.

Secondary Objective 2 – To identify relevant and discriminative features: This objective was achieved through statistical analysis, feature evaluation, and insights perceived from the existing literature that support the relevance of characteristics in DDoS detection. Features were analyzed based on their behavior, variability, and discriminative power, allowing the identification of relevant, redundant, and domain-dependent features. This objective narrows down the SO1 and was addressed throughout Chapter 4, especially in the statistical analysis and discussion sections (see Section 4.2, 4.4, 4.5, 4.6 and 4.7).

Primary Objective – To perform experimental validation using machine learning models: This objective was achieved through the application of Random Forest and SVM baseline models. Results show that the features selected in Section 4.7) were effective in distinguishing between traffic across IoT and IIoT datasets. The third objective was addressed in the experimental validation of Section 4.8.

5.4 Research Contribution

The research enhances the discussion about best practices in feature selection by providing a low-level manual feature selection, applying descriptive, exploratory, descriptive and experimental techniques.

The combination of multiple methods supports the development of evidence-based guidelines that are the greatest contribution of this study, since the analysis is not limited to one dataset only. The guidelines can support research with machine learning approach for DDoS detection by offering feature selection not limited to an specific dataset, and by stating dataset, not only the environment is taken into account, but also how the data was collected and structures, since this analysis was made with flow-based and time-window-based feature representations.

In addition, the study shows that relevance should consider a broader aspect rather than relying on one method only. As shown, not always features statistical relevance meet the relevance from existing literature.

Overall, enabling for results reproducibility is a key contribution of this project.

5.5 Future Work

Future work proposals are entirely related to the limitations found during the development of this research. The machine learning models used served the purpose of minimally experimental validation. Therefore, future work could explore more advanced machine learning approaches, including deep learning models. This would bring another objective to the research: understand how feature behavior interacts with different models.

Another limitation that became a future work proposal is performing feature engineering techniques. Despite this study relying on two different datasets, the analysis is restricted to the feature as provided in the dataset. Feature research aligned to feature engineering process could investigate whether feature creation and modification improve different models' evaluation metrics.

The last future work suggestion and final statement of this project is also a future work "wish". Future work could explore DDoS traffic over different years, allowing a cross-years analysis. This type of work could explore how patterns evolve over time, particularly if they evolve with the same rhythm as the security and mitigation mechanisms.

Overall, this study explored and demonstrated the importance of understanding feature relevance in different domains, while highlighting that feature behavior is not fixed, but dependent of dataset characteristics, specific aspects of communication patterns and traffic dynamics. The findings reinforce that the discriminative power of some features in one context might not

be the same in a different domain. Therefore, the guidelines support development of more adaptable intrusion detection systems. However, the study also proved that many factors influence feature effectiveness, such as the machine learning model selected, indicating the need for further investigation.

Finally, future research focus on extending this analysis to different datasets, models, and timeframes, contributing to more generalizable DDoS detection solutions.

References

- Adam, M., Hammoudeh, M., Alrawashdeh, R., & Alsulaimy, B. (2024). A survey on security, privacy, trust, and architectural challenges in IoT systems. *IEEE Access*, 12, 57128–57149. <https://doi.org/10.1109/ACCESS.2024.3382709>
- Al Mamun, A., Naznen, F., Yang, M., Yang, Q., Wu, M., & Masukujjaman, M. (2023). Predicting the intention and adoption of wearable payment devices using hybrid SEM-neural network analysis. *Scientific Reports*, 13(1), 11217. <https://doi.org/10.1038/s41598-023-38333-0>
- Alani, M. M., & Damiani, E. (2023). XRecon: An explainable IoT reconnaissance attack detection system based on ensemble learning. *Sensors*, 23(11), 5298. <https://doi.org/10.3390/s23115298>
- Alaoui, R. L., & Nfaoui, E. H. (2022). Deep learning for vulnerability and attack detection on web applications: A systematic literature review. *Future Internet*, 14(4), 118. <https://doi.org/10.3390/fi14040118>
- Alcock, S., & Nelson, R. (2010). An analysis of tcp maximum segment sizes.
- Ali, M. H., Jaber, M. M., Abd, S. K., Rehman, A., Awan, M. J., Damaševičius, R., & Bahaj, S. A. (2022). Threat analysis and distributed denial of service (DDoS) attack recognition in the internet of things (IoT). *Electronics*, 11(3), 494. <https://doi.org/10.3390/electronics11030494>
- Ali, Z., Bhaskar, S., & Sudheesh, K. (2019). Descriptive statistics: Measures of central tendency, dispersion, correlation and regression. *Airway*, 2(3), 120. https://doi.org/10.4103/ARWY.ARWY_37_19
- Alwaisi, Z. (2025). Memory-efficient and robust detection of mirai botnet for future 6g-enabled IoT networks. *Internet of Things*, 32, 101621. <https://doi.org/10.1016/j.iot.2025.101621>
- Aly, M., Khomh, F., Haoues, M., Quintero, A., & Yacout, S. (2019). Enforcing security in internet of things frameworks: A systematic literature review. *Internet of Things*, 6, 100050. <https://doi.org/10.1016/j.iot.2019.100050>
- Atzori, L., Iera, A., & Morabito, G. (2010). The Internet of Things: A survey. *Computer Networks*, 54(15), 2787–2805. <https://doi.org/10.1016/j.comnet.2010.05.010>
- Bali, V., Bhatnagar, V., Aggarwal, D., Bali, S., & Diván, M. J. (2021, November 4). *Cyber-physical, IoT, and autonomous systems in industry 4.0* (1st ed.). CRC Press. <https://doi.org/10.1201/9781003146711>

- Bernal, J., & Sridhar, B. (2023). *Industrial IoT for architects and engineers: Architecting secure, robust, and scalable industrial IoT solutions with AWS*. Packt Publishing.
- Bhajantri, L. B., & S., G. (2021). A comprehensive survey on resource management in internet of things. *Journal of Telecommunications and Information Technology*, 4(2020), 27–43. <https://doi.org/10.26636/jtit.2020.145220>
- Bibi, N., Iqbal, F., Akhtar, S. M., Anwar, R., & Bibi, S. (2021). A survey of application layer protocols of internet of things. *International Journal of Computer Science and Network Security*, 21(11), 301–311. <https://doi.org/10.22937/IJCSNS.2021.21.11.41>
- Bogdanoski, M., Shuminoski, T., & Risteski, A. (2013). Analysis of the SYN Flood DoS Attack. *International Journal of Computer Network and Information Security*, 5(8), 15–11. <https://doi.org/10.5815/ijcnis.2013.08.01>
- Boyes, H., Hallaq, B., Cunningham, J., & Watson, T. (2018). The industrial internet of things (IIoT): An analysis framework. *Computers in Industry*, 101, 1–12. <https://doi.org/10.1016/j.compind.2018.04.015>
- Casola, V., De Benedictis, A., Riccio, A., Rivera, D., Mallouli, W., & De Oca, E. M. (2019). A security monitoring system for internet of things. *Internet of Things*, 7, 100080. <https://doi.org/10.1016/j.iot.2019.100080>
- Chaudhary, S., & Mishra, P. K. (2023). DDoS attacks in Industrial IoT: A survey. *Computer Networks*, 236, 110015. <https://doi.org/10.1016/j.comnet.2023.110015>
- Choudhary, A. (2024). Internet of things: A comprehensive overview, architectures, applications, simulation tools, challenges and future directions. *Discover Internet of Things*, 4(1), 31. <https://doi.org/10.1007/s43926-024-00084-3>
- Cisco. (n.d.). *World IPv6 launch solidifies global support for new internet protocol*. Retrieved December 11, 2025, from <https://newsroom.cisco.com/c/r/newsroom/en/us/a/y2012/m01/world-ipv6-launch-solidifies-global-support-for-new-internet-protocol.html>
- D. Foote, K. (n.d.). *A brief history of the internet of things* [Dataversity]. Retrieved December 11, 2025, from <https://www.dataversity.net/articles/brief-history-internet-things/>
- De Donno, M., Dragoni, N., Giaretta, A., & Spognardi, A. (2018). DDoS-capable IoT malwares: Comparative analysis and mirai investigation. *Security and Communication Networks*, 2018, 1–30. <https://doi.org/10.1155/2018/7178164>
- Domínguez-Bolaño, T., Campos, O., Barral, V., Escudero, C. J., & García-Naya, J. A. (2022). An overview of IoT architectures, technologies, and existing open-source projects. *Internet of Things*, 20, 100626. <https://doi.org/10.1016/j.iot.2022.100626>
- Duc Nguyen, S., Mimura, M., & Tanaka, H. (2018). Slow-Port-Exhaustion DoS Attack on Virtual Network Using Port Address Translation. *2018 Sixth International Symposium on Computing and Networking (CANDAR)*, 126–132. <https://doi.org/10.1109/CANDAR.2018.00024>
- Ebady Manaa, M., Hussain, S. M., Suad A. Alasadi, & Hussein A. A. Al-Khamees. (2024). DDoS Attacks Detection based on Machine Learning Algorithms in IoT Environments. *Inteligencia Artificial*, 27(74), 152–165. <https://doi.org/10.4114/intartif.vol27iss74pp152-165>

- Feroz Khan, A., & G, A. (2021). A Multi-layer Security approach for DDoS detection in Internet of Things. *International Journal of Intelligent Unmanned Systems*, 9(3), 178–191. <https://doi.org/10.1108/IJIUS-06-2019-0029>
- Finsterbusch, M., Richter, C., Rocha, E., Muller, J.-A., & Hanssgen, K. (2014). A Survey of Payload-Based Traffic Classification Approaches. *IEEE Communications Surveys & Tutorials*, 16(2), 1135–1156. <https://doi.org/10.1109/SURV.2013.100613.00161>
- Firouzi, A., Dadkhah, S., Maret, S. A., & Ghorbani, A. A. (2025). DataSense: A Real-Time Sensor-Based Benchmark Dataset for Attack Analysis in IIoT with Multi-Objective Feature Selection. *Electronics*, 14(20), 4095. <https://doi.org/10.3390/electronics14204095>
- Garg, U., Verma, P., Moudgil, Y. S., & Sharma, S. (2012). Mac and logical addressing (a review study). *Int. J. Eng. Res. Appl.(IJERA)*.
- Garsva, E., Paulauskas, N., Grazulevicius, G., & Gulbinovic, L. (2014). Packet Inter-arrival Time Distribution in Academic Computer Network. *Elektronika ir Elektrotechnika*, 20(3), 87–90. <https://doi.org/10.5755/j01.eee.20.3.6683>
- Gerber, A., & Romeo, J. (2020, January 30). *IBM developer* [IBM developer]. Retrieved December 9, 2025, from <https://developer.ibm.com/articles/iot-lp101-connectivity-network-protocols/>
- Géron, A. (2018). *Hands-on machine learning with scikit-learn and TensorFlow: Concepts, tools, and techniques to build intelligent systems* (First edition, fifth release). O'Reilly.
- Griffioen, H., & Doerr, C. (2020). Examining mirai's battle over the internet of things. *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security*, 743–756. <https://doi.org/10.1145/3372297.3417277>
- Gu, T., Abhishek, A., Fu, H., Zhang, H., Basu, D., & Mohapatra, P. (2020). Towards learning-automation IoT attack detection through reinforcement learning. *2020 IEEE 21st International Symposium on "A World of Wireless, Mobile and Multimedia Networks" (WoWMoM)*, 88–97. <https://doi.org/10.1109/WoWMoM49955.2020.00029>
- Gvozdenovic, S., Becker, J. K., Mikulskis, J., & Starobinski, D. (2024). IoT-scan: Network reconnaissance for internet of things. *IEEE Internet of Things Journal*, 11(8), 13091–13107. <https://doi.org/10.1109/JIOT.2023.3327293>
- HaddadPajouh, H., Dehghantanha, A., M. Parizi, R., Aledhari, M., & Karimipour, H. (2021). A survey on internet of things security: Requirements, challenges, and solutions. *Internet of Things*, 14, 100129. <https://doi.org/10.1016/j.iot.2019.100129>
- Internet Protocol. (1981, September). <https://doi.org/10.17487/RFC0791>
- IPv4Mall. (2024, July 7). *IPv4 and IoT: Addressing challenges and opportunities* [IPv4mall]. Retrieved November 27, 2025, from <https://blogs.ipv4mall.com/blogs/ipv4-and-iot-addressing-challenges-and-opportunities/>
- Jerry Sheehan. (2025, May). Understanding the TCP PSH Flag in Networking. https://synchronet.net/tcp-psh-flag/#PSH_vs_ACK_Understanding_Key_Differences
- Jo, I., Lee, S., & Oh, S. (2019). Improved Measures of Redundancy and Relevance for mRMR Feature Selection. *Computers*, 8(2), 42. <https://doi.org/10.3390/computers8020042>

- Jullian, O., Otero, B., Rodriguez, E., Gutierrez, N., Antona, H., & Canal, R. (2023). Deep-learning based detection for cyber-attacks in IoT networks: A distributed attack detection framework. *Journal of Network and Systems Management*, 31(2), 33. <https://doi.org/10.1007/s10922-023-09722-7>
- Kadri, M. R., Abdelli, A., Ben Othman, J., & Mokdad, L. (2024). Survey and classification of dos and DDos attack detection and validation approaches for IoT environments. *Internet of Things*, 25, 101021. <https://doi.org/10.1016/j.iot.2023.101021>
- Karimi, M., & Shaefer, R. (2025, March 25). IIoT communication protocols compatibility and security an in-depth review. <https://doi.org/10.36227/techrxiv.174286607.78655824/v1>
- Kayode Saheed, Y., Idris Abiodun, A., Misra, S., Kristiansen Holone, M., & Colomo-Palacios, R. (2022). A machine learning-based intrusion detection for detecting internet of things network attacks. *Alexandria Engineering Journal*, 61(12), 9395–9409. <https://doi.org/10.1016/j.aej.2022.02.063>
- Khalil, M. (2025, August 24). *IoT hacking statistics 2025: Threats, risks & regulations* [Deep-Strike]. Retrieved December 9, 2025, from <https://deepstrike.io/blog/iot-hacking-statistics>
- Khan, F., Al-Atawi, A. A., Alomari, A., Alsirhani, A., Alshahrani, M. M., Khan, J., & Lee, Y. (2022). Development of a model for spoofing attacks in internet of things. *Mathematics*, 10(19), 3686. <https://doi.org/10.3390/math10193686>
- Khujamatov, H., Reypnazarov, E., Khasanov, D., & Akhmedov, N. (2021). IoT, IIoT, and cyber-physical systems integration. In K. K. Singh, A. Nayyar, S. Tanwar, & M. Abouhawwash (Eds.), *Emergence of cyber physical system and IoT in smart automation and robotics* (pp. 31–50). Springer International Publishing. https://doi.org/10.1007/978-3-030-66222-6_3
- Kumar Mishra, A., & Raghuvanshi, C. S. (2024, February 29). FUTURISTIC TRENDS IN IoT. In P. (Ipseeta Nanda, D. F. A. Mazarbhuiya, D. N. Hussain, D. E. Channaveeramma, D. N. Agarwal, D. V. Kumara, A. Mattoo, & D. Singhal (Eds.), *Futuristic trends in IOT volume 3 book 3* (First, pp. 65–108). Iterative International Publishers, Selfypage Developers Pvt Ltd. <https://doi.org/10.58532/V3BDIO3P1CH4>
- Latorre, L., Rego, E., & De Leo, L. (2025, February 26). *Tech report: Internet of things*. Inter-American Development Bank. <https://doi.org/10.18235/0013422>
- Leng, J., Chen, Z., Huang, Z., Zhu, X., Su, H., Lin, Z., & Zhang, D. (2022). Secure blockchain middleware for decentralized IIoT towards industry 5.0: A review of architecture, enablers, challenges, and directions. *Machines*, 10(10), 858. <https://doi.org/10.3390/machines10100858>
- Li, S. (2017). *Securing the internet of things*. Elsevier Science.
- Luque, A., Carrasco, A., Martín, A., & De Las Heras, A. (2019). The impact of class imbalance in classification performance metrics based on the binary confusion matrix. *Pattern Recognition*, 91, 216–231. <https://doi.org/10.1016/j.patcog.2019.02.023>

- McKinsey. (2018, May). *The internet of things: How to capture the value of IoT*. McKinsey. <https://www.mckinsey.com/~media/mckinsey/business%20functions/mckinsey%20digital/our%20insights/the%20internet%20of%20things%20how%20to%20capture%20the%20value%20of%20iot/how-to-capture-the-value-of-iot.pdf>
- Mgungile, J., & Tonkal, Ö. (2025). Scalable Intrusion Detection in IoT Networks: A Big Data Analytics Approach. *Turkish Journal of Engineering*, 10(1), 230–243. <https://doi.org/10.31127/tuje.1793847>
- Mirani, A. A., Velasco-Hernandez, G., Awasthi, A., & Walsh, J. (2022). Key challenges and emerging technologies in industrial IoT architectures: A review. *Sensors*, 22(15), 5836. <https://doi.org/10.3390/s22155836>
- Munirathinam, S. (2020, January 1). Industry 4.0: Industrial internet of things (IIOT). In *Advances in computers* (pp. 129–164, Vol. 117). Elsevier. <https://doi.org/10.1016/bs.adcom.2019.10.010>
- Munshi, A. (2024). Detecting Unusual Repetitive Patterns of Behavior Indicative of a Loop-Based Attack in IoT. *Sensors*, 24(23), 7534. <https://doi.org/10.3390/s24237534>
- Neto, E. C. P., Dadkhah, S., Ferreira, R., Zohourian, A., Lu, R., & Ghorbani, A. A. (2023). CICIoT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment. *Sensors*, 23(13), 5941. <https://doi.org/10.3390/s23135941>
- Netscout's. (n.d.). *NETSCOUT threat intelligence report* [Latest cyber threat intelligence report]. Retrieved December 9, 2025, from <https://www.netscout.com/threatreport/>
- Niakanlahiji, A., Pritom, M. M., Chu, B.-T., & Al-Shaer, E. (2017). Predicting Zero-day Malicious IP Addresses. *Proceedings of the 2017 Workshop on Automated Decision Making for Active Cyber Defense*, 1–6. <https://doi.org/10.1145/3140368.3140369>
- Otoom, A. F., Eleisah, W., & Abdallah, E. E. (2023). Deep learning for accurate detection of brute force attacks on IoT networks. *Procedia Computer Science*, 220, 291–298. <https://doi.org/10.1016/j.procs.2023.03.038>
- Pakmehr, A., Aßmuth, A., Taheri, N., & Ghaffari, A. (2024). DDoS attack detection techniques in IoT networks: A survey. *Cluster Computing*, 27(10), 14637–14668. <https://doi.org/10.1007/s10586-024-04662-6>
- Palla, T. G., & Tayeb, S. (2021). Intelligent mirai malware detection for IoT nodes. *Electronics*, 10(11), 1241. <https://doi.org/10.3390/electronics10111241>
- Peter, O., Pradhan, A., & Mbohwa, C. (2023). Industrial internet of things (IIoT): Opportunities, challenges, and requirements in manufacturing businesses in emerging economies. *Procedia Computer Science*, 217, 856–865. <https://doi.org/10.1016/j.procs.2022.12.282>
- Petroc, T. (2025, November 19). *IoT connections worldwide 2034* [Statista]. Retrieved December 11, 2025, from <https://www.statista.com/statistics/1183457/iot-connected-devices-worldwide/?srsltid=AfmBOopkjXb-oMkzKmCYe8l6ebN5XksQC7s15YN4W-1RsMtPhz8mobjY>

- Pilcher, N., & Cortazzi, M. (2024). 'qualitative' and 'quantitative' methods and approaches across subject fields: Implications for research values, assumptions, and practices. *Quality & Quantity*, 58(3), 2357–2387. <https://doi.org/10.1007/s11135-023-01734-4>
- Rabbani, M., Gui, J., Nejati, F., Zhou, Z., Kaniyamattam, A., Mirani, M., Piya, G., Opushnyev, I., Lu, R., & Ghorbani, A. A. (2025). Device Identification and Anomaly Detection in IoT Environments. *IEEE Internet of Things Journal*, 12(10), 13625–13643. <https://doi.org/10.1109/JIOT.2024.3522863>
- Rehan, H. (2023). Internet of things (IoT) in smart cities: Enhancing urban living through technology. *Journal of Engineering and Technology*, 5(1), 116. Retrieved November 15, 2025, from <https://mzresearch.com/index.php/JET/article/view/70>
- Roopak, M., Tian, G. Y., & Chambers, J. (2020). Multi-objective-based feature selection for DDoS attack detection in IoT networks. *IET Networks*, 9(3), 120–127. <https://doi.org/10.1049/iet-net.2018.5206>
- Rui, H. Z., Chien, T. Y., Ee, L. X., Wei San, C., & Teng Yi, L. (2025). Comparison of the Use of Support Vector Machine (SVM) & Random Forest Algorithms (RF) for DDOS Attack Detection. *International Journal of Research and Innovation in Social Science*, 9(1), 1126–1138. <https://doi.org/10.47772/IJRISS.2025.9010094>
- Sadique, K. M., Rahmani, R., & Johannesson, P. (2018). Towards security on internet of things: Applications and challenges in technology. *Procedia Computer Science*, 141, 199–206. <https://doi.org/10.1016/j.procs.2018.10.168>
- Sahu, A. K., Sharma, S., Tanveer, M., & Raja, R. (2021). Internet of things attack detection using hybrid deep learning model. *Computer Communications*, 176, 146–154. <https://doi.org/10.1016/j.comcom.2021.05.024>
- Salman, T., & Jain, R. (2019). A survey of protocols and standards for internet of things. <https://doi.org/10.48550/ARXIV.1903.11549>
- Salvatore S. Mangiafico. (2026). *SUMMARY AND ANALYSIS OF EXTENSION PROGRAM EVALUATION IN R* (1.24.4). Rutgers Cooperative Extension New Brunswick.
- Sayed, M. S. E., Le-Khac, N.-A., Azer, M. A., & Jurcut, A. D. (2022). A Flow-Based Anomaly Detection Approach With Feature Selection Method Against DDoS Attacks in SDNs. *IEEE Transactions on Cognitive Communications and Networking*, 8(4), 1862–1880. <https://doi.org/10.1109/TCCN.2022.3186331>
- Sharmin, N., & Kiekintveld, C. (2023). Enhancing IoT device security: Predicting and analyzing reconnaissance attacks using flags and time-based attributes. *2023 10th International Conference on Internet of Things: Systems, Management and Security (IOTSMS)*, 23–30. <https://doi.org/10.1109/IOTSMS59855.2023.10325811>
- Singh, C., & Jain, A. K. (2025). Defense strategies of DDoS attacks in SDN-IoT network: A survey. *Procedia Computer Science*, 259, 809–817. <https://doi.org/10.1016/j.procs.2025.04.032>
- Sinha, S. (2025). *Number of connected IoT devices growing 14% to 21.1 billion*. Retrieved November 10, 2025, from <https://iot-analytics.com/number-connected-iot-devices/>

- Sisinni, E., Saifullah, A., Han, S., Jennehag, U., & Gidlund, M. (2018). Industrial Internet of Things: Challenges, Opportunities, and Directions. *IEEE Transactions on Industrial Informatics*, 14(11), 4724–4734. <https://doi.org/10.1109/TII.2018.2852491>
- Snyder, H. (2019). Literature review as a research methodology: An overview and guidelines. *Journal of Business Research*, 104, 333–339. <https://doi.org/10.1016/j.jbusres.2019.07.039>
- Stiawan, D., Idris, M. Y., Malik, R. F., Nurmaini, S., Alsharif, N., & Budiarto, R. (2019). Investigating brute force attack patterns in IoT network. *Journal of Electrical and Computer Engineering*, 2019, 1–13. <https://doi.org/10.1155/2019/4568368>
- Talal, H., & Zagrouba, R. (2021). MADS based on DL techniques on the internet of things (IoT): Survey. *Electronics*, 10(21), 2598. <https://doi.org/10.3390/electronics10212598>
- Trifonov, H., & Heffernan, D. (2023). OPC UA TSN: A next-generation network for industry 4.0 and IIoT. *International Journal of Pervasive Computing and Communications*, 19(3), 386–411. <https://doi.org/10.1108/IJPCC-07-2021-0160>
- Vinicius Fulber-Garcia & Michal Aibin. (2024, March). TCP: Differences Between FIN and RST. <https://www.baeldung.com/cs/tcp-fin-vs-rst>
- Xu, L. D., He, W., & Li, S. (2014). Internet of Things in Industries: A Survey. *IEEE Transactions on Industrial Informatics*, 10(4), 2233–2243. <https://doi.org/10.1109/TII.2014.2300753>
- Zhang, X., Upton, O., Beebe, N. L., & Choo, K.-K. R. (2020). IoT botnet forensics: A comprehensive digital forensic case study on mirai botnet servers. *Forensic Science International: Digital Investigation*, 32, 300926. <https://doi.org/10.1016/j.fsidi.2020.300926>
- Zhao, K., & Ge, L. (2013). ASurvey on the internet of things security. *2013 Ninth International Conference on Computational Intelligence and Security*.
- Zhou, L., Liao, M., Yuan, C., & Zhang, H. (2017). Low-Rate DDoS Attack Detection Using Expectation of Packet Size. *Security and Communication Networks*, 2017, 1–14. <https://doi.org/10.1155/2017/3691629>
- Zughrat, A., Mahfouf, M., Yang, Y., & Thornton, S. (2014). Support Vector Machines for Class Imbalance Rail Data Classification with Bootstrapping-based Over-Sampling and Under-Sampling. *IFAC Proceedings Volumes*, 47(3), 8756–8761. <https://doi.org/10.3182/20140824-6-ZA-1003.00794>



Sample Appendix

The source code, data preprocessing, statistical analysis and machine learning implementation used in this project are available at: https://github.com/ecsouzam/ADS_bachelor_project

Word count metrics

NUC Bachelor Project Word Count:

Total Sum count: 20661 Words in text: 20187 Words in headers: 295 Words outside text (captions, etc.): 73 Number of headers: 98 Number of floats/tables/figures: 9 Number of math inlines: 106 Number of math displayed: 0 (errors:2) NOTE: References are excluded.